LUANGWA COMMUNITY FORESTS PROJECT

Document Prepared By BioCarbon Partners

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Table of Contents

List of VM0009 3.0 MR requirements applicable to F-U3 baseline type (See VM0009 Appendix I).

MR#	Category	Applicable to Project	Page#
MR.1	Spatial Project Boundaries	Yes	17
MR.2	Temporal Project Boundaries	Yes	15
MR.3	Temporal Project Boundaries	Yes	15
MR.4	Grouped Projects	Yes	15
MR.5	Grouped Projects	Yes	17
MR.6	Project Accounting Areas	Yes	17
MR.7	Determining t_{PAI}	Yes	38
MR.8	Determining t_{PAI}	Yes	38
MR.9	Determining x_{PAI}	No. No covarites to conversion were used.	
MR.10	Baseline Emissions	Yes	83
MR.11	Baseline Emissions	Yes	83
MR.12	Baseline Emissions	Yes	84
MR.15	Applying the Spatial Algorithm	Yes, but there is only one stratum	84
MR.16	Applying the Spatial Algorithm	Yes, but there is only one stratum	84
MR.17	Applying the Spatial Algorithm	Yes, but there is only one stratum	84
MR.21	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios	
MR.22	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios	
MR.23	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios	
MR.24	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios	
MR.25	Carbon Not Decayed in DW	No. All biomass is burned in baseline scenarios	
MR.26	Carbon Not Decayed in BGB	Yes	84
MR.27	Carbon Not Decayed in BGB	Yes	84
MR.28	Carbon Not Decayed in BGB	Yes	84
MR.29	Carbon Not Decayed in SOC	Yes	85
MR.30	Carbon Stored in Wood Products	No. No timber harvesting in baseline	
MR.31	Carbon Stored in Wood Products	No. No timber harvesting in baseline	

MR.32	Emissions Events in Project Area	Yes	85
MR.33	Emissions Events in Project Area	Yes	86
MR.34	Emissions from Burning from Project Activities	No. No biomass burning in project activities	
MR.35	Carbon Stored in Wood Products from Project Activities	No. Timber harvesting is not a project activity	
MR.36	Carbon Stored in Wood Products from Project Activities	No. Timber harvesting is not a project activity	
MR.37	Carbon Stored in Wood Products from Project Activities	No. Timber harvesting is not a project activity	
MR.38	Livestock Grazed in the Project Area	No. Livestock grazing is not permitted in the project area and unpermited grazing was determined to be de minimus.	
MR.39	Livestock Grazed in the Project Area	No. Livestock grazing is not permitted in the project area and unpermited grazing was determined to be de minimus.	
MR.40	Livestock Grazed in the Project Area	No. Livestock grazing is not permitted in the project area and unpermited grazing was determined to be de minimus.	
MR.41	Synthetic Fertilizer in the Project Area	No. Project activities do not include the use of synthetic fertilizers.	
MR.42	Synthetic Fertilizer in the Project Area	No. Project activities do not include the use of synthetic fertilizers.	
MR.43	Synthetic Fertilizer in the Project Area	No. Project activities do not include the use of synthetic fertilizers.	
MR.44	Leakage Mitigation Strategies	Yes	91
MR.45	Commodity Production for Leakage Mitigation	No. The baseline scenario is subsistence	
MR.46	Commodity Production for Leakage Mitigation	No. The baseline scenario is subsistence	
MR.47	Commodity Production for Leakage Mitigation	No. The baseline scenario is subsistence	
MR.48	Estimating Emissions from Activity-Shifting Leakage	Yes	91
MR.49	Estimating Emissions from	Yes	91

	Activity-Shifting Leakage		
MR.50	Change to the Activity-Shifting Leakage Area	No. No change to activity-shifting leakage area	
MR.51	Change to the Activity-Shifting Leakage Area	No. No change to activity-shifting leakage area	
MR.52	Change to the Activity-Shifting Leakage Area	No. No change to activity-shifting leakage area	
MR.53	Change to the Activity-Shifting Leakage Area	No. No change to activity-shifting leakage area	
MR.54	Change to the Activity-Shifting Leakage Area	No. No change to activity-shifting leakage area	
MR.55	Change to the Activity-Shifting Leakage Area	No. No change to activity-shifting leakage area	
MR.56	Estimating $p_{L\ DEG}$	Yes	91
MR.57	Estimating $p_{L\ DEG}$	Yes	91
MR.58	Estimating $p_{L\ CONG}$	No. No grassland in project area	
MR.59	Estimating $p_{L\ CONG}$	No. No grassland in project area	
MR.60	Determining Emissions from Market Leakage	No. The baseline scenario is subsistence	
MR.61	Determining Emissions from Market Leakage	No. The baseline scenario is subsistence	
MR.62	Determining Emissions from Market Leakage	No. The baseline scenario is subsistence	
MR.63	Ensuring No Leakage Within Project Proponent's Ownership	Yes	91
MR.65	Quantification of GERs	Yes	92
MR.66	Quantification of GERs	Yes	92
MR.67	Quantification of GERs	Yes	92
MR.68	Confidence Deduction	Yes	93
MR.69	Confidence Deduction	Yes	93
MR.72	Reversal Event	No. No reversal occurred	

MR.73	Reversal Event as a Result of Baseline Reevaluation	No. No baseline reevaluation occurred	
MR.74	Quantification of NERs for a PAA	Yes	93
MR.75	Quantification of NERs for a PAA	Yes	94
MR.76	Quantification of NERs for a PAA	Yes	94
MR.77	Buffer Account	Yes	94
MR.78	Buffer Account	Yes	94
MR.79	Quantification of NERs across PAAs	No. There is currently only 1 PAA	
MR.80	Quantification of NERs across PAAs	No. There is currently only 1 PAA	
MR.81	Quantification of NERs across PAAs	No. There is currently only 1 PAA	
MR.82	Vintages	Yes	95
MR.83	Evaluating Project Performance	Yes	95
MR.84	Evaluating Project Performance	Yes	95
MR.85	Data and Parameters Monitored	Yes	49
MR.86	Data and Parameters Monitored	Yes	49
MR.87	Data and Parameters Monitored	Yes	49
MR.88	Description of the Monitoring Plan	Yes	79
MR.89	Description of the Monitoring Plan	No. No project activity with biomass burning	
MR.90	Description of the Monitoring Plan	Yes	79
MR.91	Description of the Monitoring Plan	Yes	76
MR.92	Description of the Monitoring Plan	Yes	77
MR.93	Description of the Monitoring Plan	Yes	75
MR.94	Description of the Monitoring Plan	No. No new equations were developed	
MR.95	Description of the Monitoring Plan	Yes	83
MR.96	Description of the Monitoring Plan	No. Timber harvesting is not a project activity	
MR.97	Description of the Monitoring Plan	Yes	36
MR.98	Description of the Monitoring Plan	Yes	77

MR.99	Sources of Allometry	Yes	80
MR.100	Sources of Allometry	Yes	80
MR.101	Sources of Allometry	Yes	81
MR.102	Sources of Allometry	Yes	81
MR.103	Sources of Allometry	Yes	81
MR.104	Validating Previously Developed Allometry	Yes	81
MR.105	Validating Previously Developed Allometry	Yes	81
MR.106	Validating Previously Developed Allometry	Yes	82
MR.107	Validating Previously Developed Allometry	Yes	82
MR.108	Validating Previously Developed Allometry	Yes	82
MR.109	Validating Newly Developed Allometry	No. No new allometric equations were developed.	
MR.110	Validating Newly Developed Allometry	No. No new allometric equations were developed.	
MR.111	Validating Newly Developed Allometry	No. No new allometric equations were developed.	
MR.112	Validating Newly Developed Allometry	No. No new allometric equations were developed.	
MR.113	Validating Newly Developed Allometry	No. No new allometric equations were developed.	

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Luangwa Community Forests Project is a large scale REDD+ project implemented in Eastern and Lusaka Province, Zambia with initial project area of 943,676 ha (Figure 1). It is being implemented on communal land in 12 chiefdoms falling within Game Management Areas (GMA) and two private ranches. Implementation is in partnership with the traditional authorities and the government of the Republic of Zambia. The project will generate emissions reductions through avoided deforestation, using the following mitigation activities: a combination of direct conservation support (forest monitoring and encroachment prevention) engagement and capacity building with key Government and community stakeholders, and conservation incentives for the area protected: including performance-based payments delivered to empowered community stakeholders, through local institutions, and support to deforestation mitigation activities, including sustainable, improved livelihoods activities, such as conservation agriculture, non-timber forest products, sustainable enterprise development, infrastructure development and water provision. Forest monitoring will be done using remote sensing, aerial and ground monitoring.

The baseline scenario is ongoing unplanned deforestation driven primarily by subsistence agriculture. Furthermore communities remain in an impoverished state and biodiversity declines due to habitat loss and increased poaching.

The project's community objective is poverty alleviation activities benefiting at least 10,000 households, specifically targeting vulnerable households and the poorest of the poor. The biodiversity objectives is maintaining a massive wildlife corridor between several national parks in the catchment of Zambia's 4th largest river system. Conserving and maintaining vulnerable and endangered species through habitat protection and reduction in poaching. The climate objectives are to avoid deforestation in the project area and assist communities and biodiversity with climate change adaptation benefits through income diversification, improved farming techniques, crop diversification and maintenance of habitat corridors.

The estimated emissions reduction for the 30 year project span is 83,598,204 tCO2e at an average 2,985,650 tCO2e/yr. The emissions reduction for the current monitoring period (May 1st, 2015 to Dec. 31st, 2018) is 3,535,625 tCO2e.

1.2 Sectoral Scope and Project Type

The project falls under scope 14 of the VCS, namely Agriculture, Forestry and Other Land Use (AFOLU). The AFOLU project category is Reducing Emissions from Deforestation and Forest Degradation.

1.3 Project Proponent

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1.4 Other Entities Involved in the Project

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Organization name	Ministry of Agriculture
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Organization name	Ministry of Agriculture
Role in the project	Agricultural Extension Support

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1.5 Project Start Date

MR.2

The project start date is May 1st, 2015.

1.6 Project Crediting Period

MR.3

The project crediting period is from May 1st, 2015 to April 30th, 2045 for a total of 30 years.

1.7 Project Location

The 14 first activity instances (Table 1, Table 2) lie in the Eastern and Lusaka Provinces of Zambia and fall into the Luangwa Zambezi River ecological zone and the Central Southern and Eastern Plateau Zone (Figure 1). In addition all the activity instances are situated in the Game Management areas (GMA) within the Chiefdom areas. The project sites in Eastern Province are border by the South Luangwa National Park and Luangwa River in the north-western direction. The project area consist of two sites is Lusaka Province in Rufunsa District, one in each side of the great east road in Shikabeta and Mpanshya chiefdoms. The others nine project sites lies in eastern province of Zambia in Mambwe, Lundazi and Nyimba Districts. Project sites in Mambwe Districts are in Mnkhanya, Jumbe, Nsefu, Malama, Msoro. In Lundazi District there is one project site in Mwanya Chiefdom. The project area in Mwanya is bordered by Lukusuzi National Park and Luambe National Park area. There are two sites in Nyimba District one each in Nyalugwe and Luembe chiefdom. The full description of the boundaries of each site can be found in the attached narratives. Below is the location map of the project area. The geographic boundaries of the project area is provided in a separate kml file.

MR.4

Table 1. Summary of Project Area - REDD+ Project Zones under communal tenure, first activity instances.

Community Resource Board	Game Management Area	Chiefdom	Size (ha)	Forest Area (ha)

Mpanshya	Rufunsa	Mpanshya	84,530	82,183
Shikabeta	Luano	Shikabeta	118,102	111,968
Mwanya	Lower Lumimba	Mwanya	81,668	59,030
Luembe	West Petauke	Luembe	288,726	254,582
Nyalugwe	West Petauke	Nyalugwe	60,893	58,809
Msoro	Lower Lupande	Msoro	24,688	23,523
Malama	Lower Lupande	Malama	21,577	17,220
Mnkhanya	Upper Lupande	Mnkhanya	33,623	29,081
Jumbe	Upper Lupande	Jumbe	15,443	11,066
Nsefu	Upper Lupande	Nsefu	39,801	25,341
Chitungulu	Lumimba	Chitungulu	38,764	28,681
Sandwe	Sandwe	Sandwe	92,362	85,767
Total			900,177	787,251

Table 2. Summary of Project Area - REDD+ Project Zones on Private Land, first activity instances.

Private Game Ranch	Chiefdom	Size (ha)	Forest Area (ha)
Munyamadzi	Luembe	18,671	16,449
Pia Manzi	Nyalugwe	24,798	22,393
Total		43,469	38,842

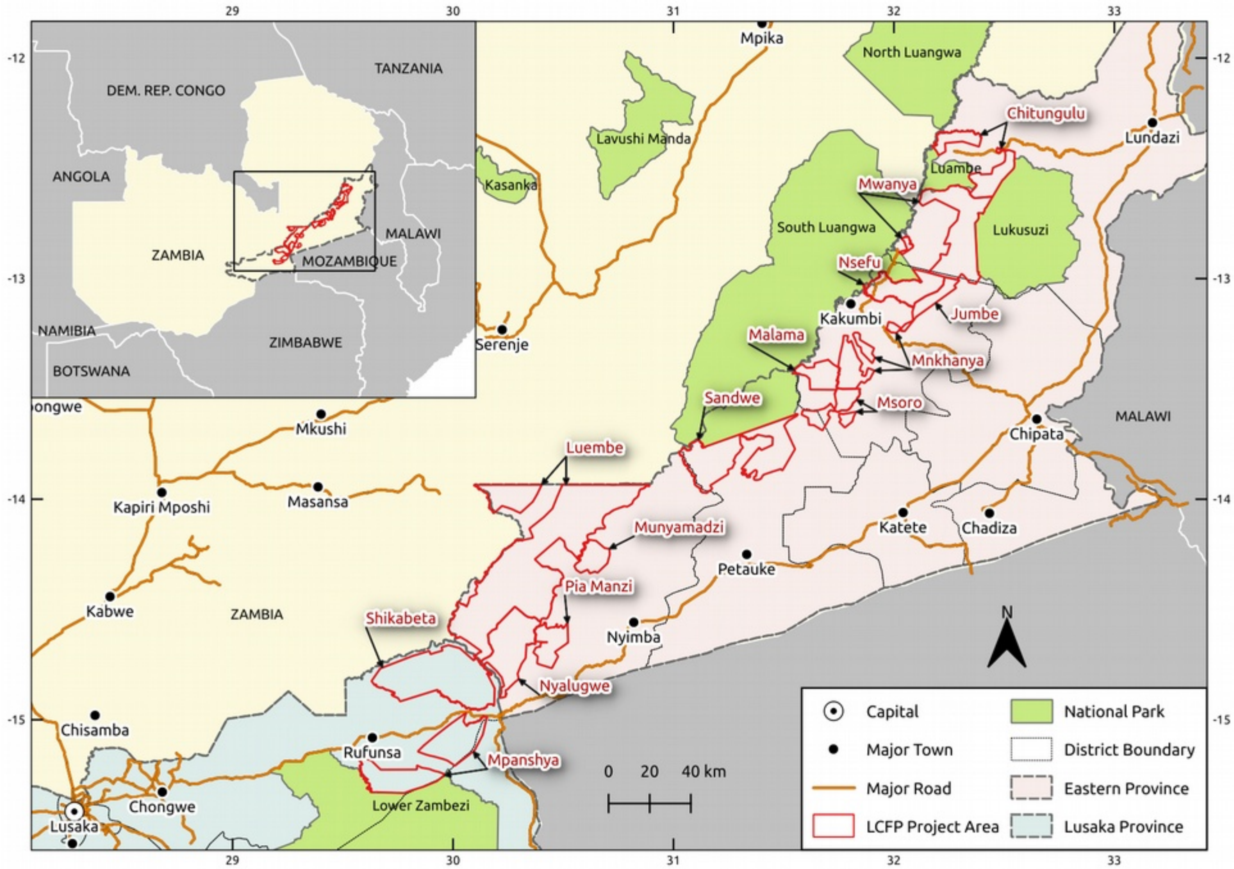

Figure 1: LCFP REDD+ initial project areas (**MR.1, MR.5**).

MR.6

As described in detail in section 1.1.2 of the PD, a GIS analysis of deforestation map from 2007 to 2017 was used to identify key constraints to conversion. Two factors stood out most prominently- distance from roads and slope. These factors make sense since roads provide access to forested areas for farmers and charcoal makers and farmers, who are responsible for most deforestation, prefer to cultivate on relatively flat ground which will better hold soil moisture and is easier to work with oxen (which some farmers have). The analysis found constraints to conversion at greater than 45% slope and greater than 6.5 km distance from a road. Therefore, to calculate the project accounting area, a polygon buffer covering 6.5 km from roads was clipped from the project area. Then, areas of high slope and areas that were not forest according to the PALSAR analysis in both 2007[1] and 2015, were also removed from the accounting area polygon.

Maps of the accounting area with topography (areas great than 45% slope), roads, major rivers and perennial streams, land-use (forest map) and accounting area boundaries are shown in Figure 2, Figure 3, Figure 4, and Figure 5.

1 This deviation from the methodology is explained and justified in section 2.2 of this report.

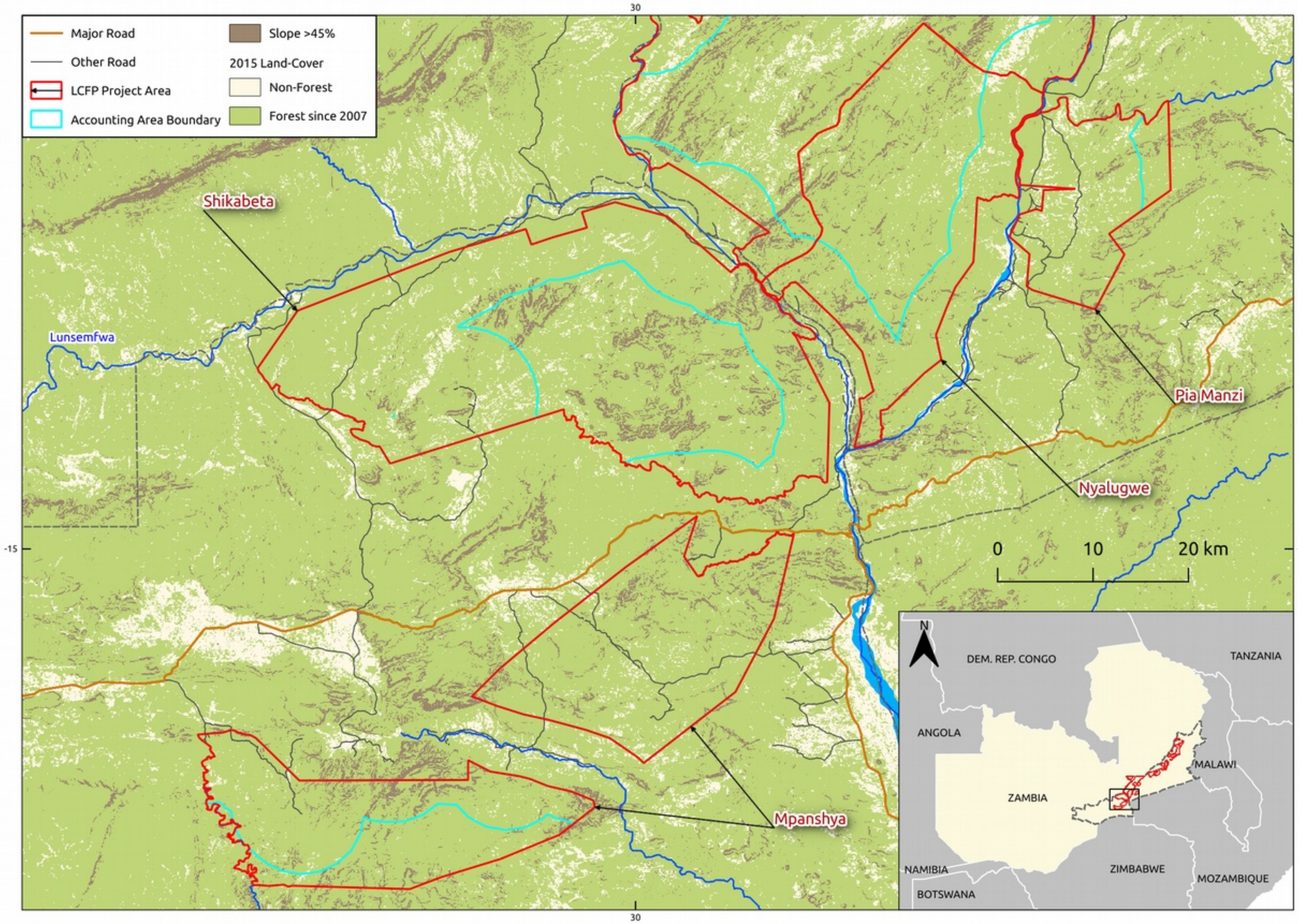

Figure 2: Mpanshya, Shikabeta, Nyalugwe, and Pia Manzi Accounting Areas

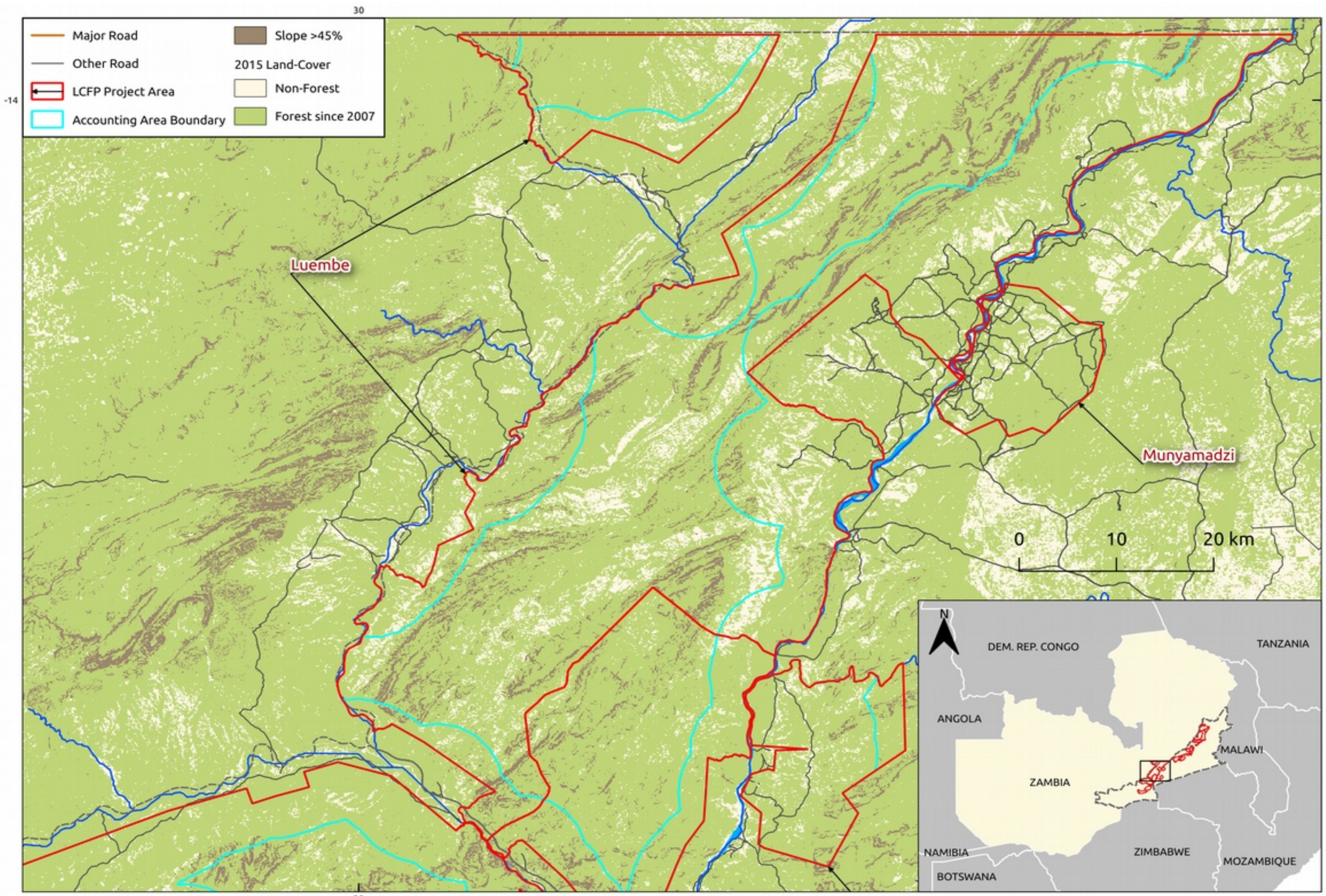

Figure 3: Luembe and Munyamadzi Accounting Areas

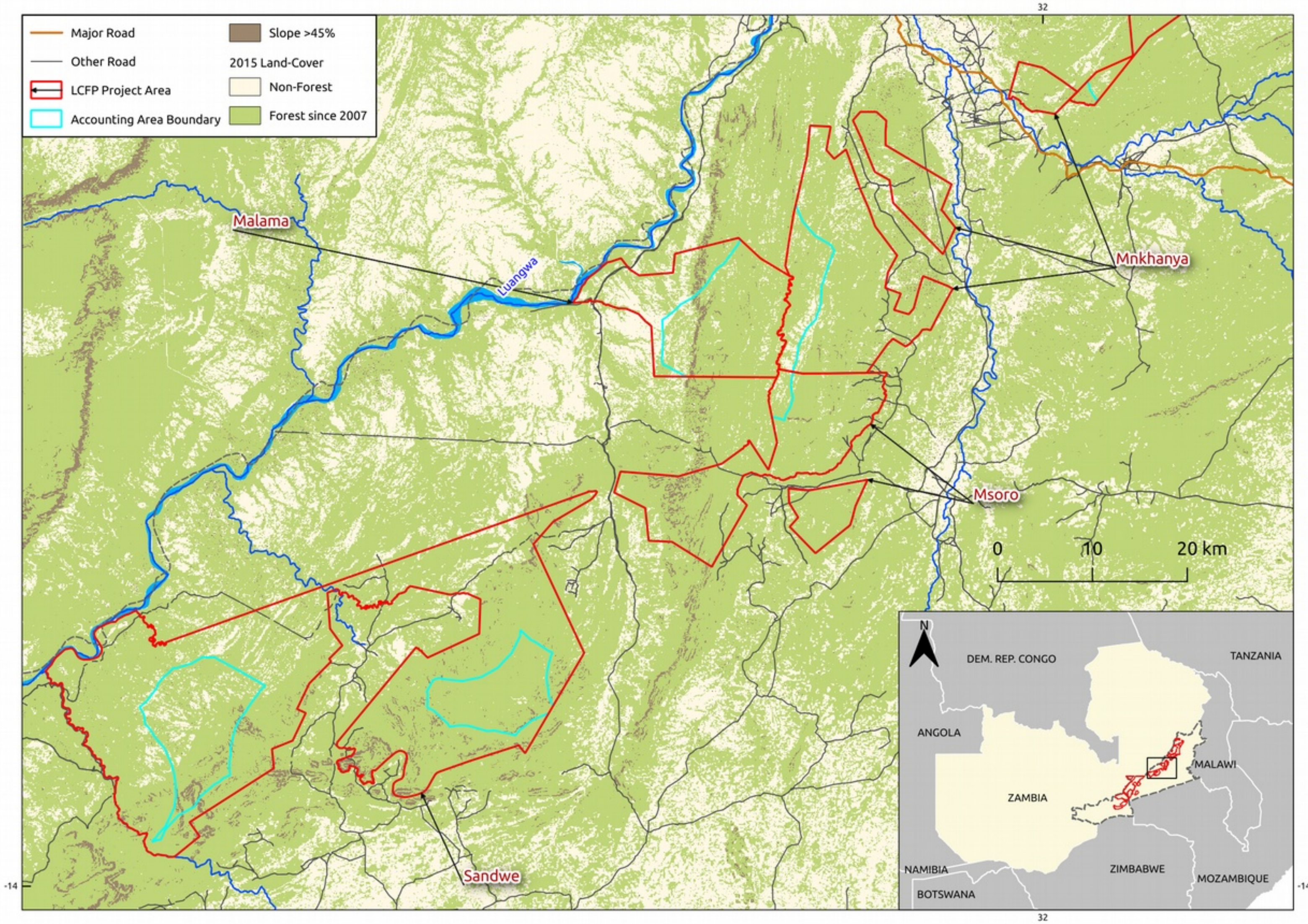

Figure 4: Sandwe, Malama, Msoro, and Mnkhanya Accounting Areas

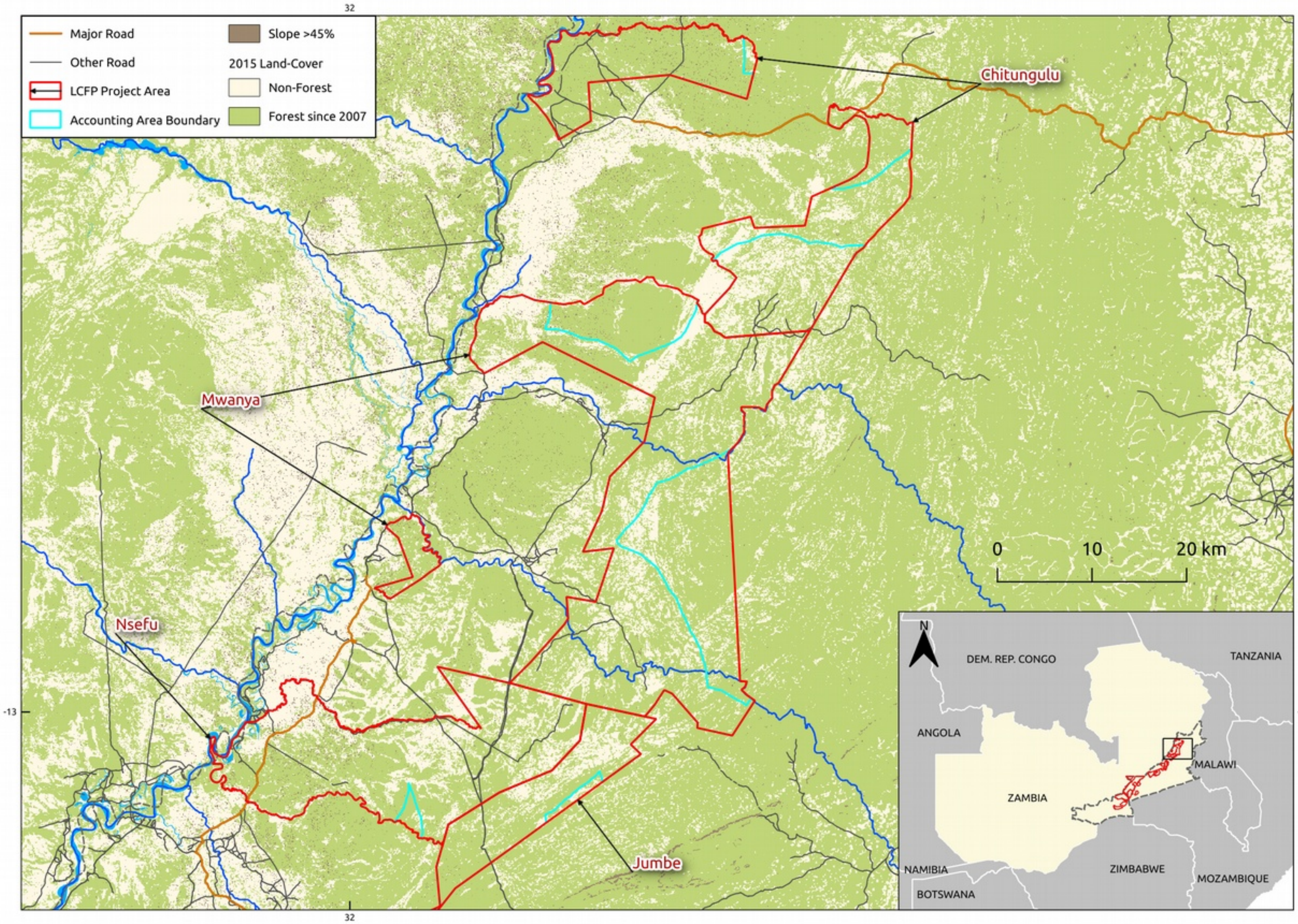

Figure 5: Jumbe, Nsefu, Myanya, and Chitulungu Accounting Areas

1.8 Title and Reference of Methodology

VCS VM0009 V3.0 Methodology for Avoided Ecosystem Conversion

VCS VT0001 Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities, version 3.0

VCS AFOLU Non-Permanence Risk Tool v3.3

1.9 Other Programs

Emissions trading and other binding limits:

The LCFP is not subject to binding emissions limits or emissions trading programmes.

Participation under other GHG programs:

The project has not applied to any other GHG programmes.

Other forms of environmental credit:

The project is seeking validation under the CCB Standard Version 3.

1.10 Sustainable Development

Nationally stated sustainable development priorities in Zambia are described and monitored through 5 year national development plans. The focus of the Government of Zambia's Five-Year 7th National Development Plan (2017-2021) is on targeting economic diversification and job creation; poverty and vulnerability reduction; reducing developmental inequalities; enhancing human development; and the creation of a conducive governance for a diversified economy. It sets targets for bringing additional areas of forest under sustainable management (500,000 hectares) and emphasizes measures to enhance the viability and diversification of agriculture co-operatives, farmer groups and community forest management groups.

The 7th NDP departs from sectoral-based planning to an integrated (multi-sectoral) development approach, which recognises the multi-faceted and interlinked nature of sustainable development. It calls for interventions to be tackled simultaneously through a coordinated approach to implementing development interventions, through clusters – involving all arms of Government, Cooperating Partners, Private Sectors, Civil Society Organisations, Faith Based Organisation and other interest groups. Further, the 7th NDP will be primarily implemented through programmes and projects that are devolved to the districts within the tenets of decentralised planning, monitoring and evaluation as determined by municipal and district councils. This will be done with a view to deal with district-specific development challenges.

The Districts design their own plans and monitoring and evaluation systems that will promote integration of development priorities at local level to provincial and national strategic areas of focus and outcomes.

In terms of supporting implementation of the 7th NDP – the five-year target of 500,000 hectares of forest under management increased will be exceeded just through the approval of the forest management agreements by the Director of Forestry under the initial BCP supported initiatives (943,676 ha) by just under a factor of 2. Reporting and monitoring on the LCFP contribution to the implementation of the 7th NDP is supporting at various levels. BCP is a member of the District Development Coordinating Committee which is chaired by the District Commissioner and Council Secretary, in each of the Districts LCFP operates. However, through our close partnership with Government Department and Agencies in each district sustainable development results and achievements are reported through their respective representatives. This includes: Department of National Parks and Wildlife, Forestry Department, Ministry of Chiefs and Traditional Affairs, Ministry of Agriculture, Ministry of Community Development and Social Welfare. At National level, BCP provides a presence on various national technical advisory committees and direct reporting as required to the line Ministries and their Departments. BCP contributes information and data to the National REDD+ Registry and Database Management System maintained by the Forestry Department. This allows Government of Zambia to monitor REDD+ activities covering information such as Project Documents, budgets, project area datasets, Certification Standards, emission reduction quantification and MRV, benefit sharing and safeguards.

2 IMPLEMENTATION STATUS

2.1 Implementation Status of the Project Activity

The LCFP has developed a wide range of deforestation mitigation and benefit sharing activities with local stakeholders. These activities fit broadly into three different categories namely infrastructure support and development, improved livelihoods and income diversification through value addition to non-timber forest products and thirdly forest encroachment and wildlife monitoring and poaching reduction. Project activities happen in the project area, project zone, and leakage areas shown in Figure 6.

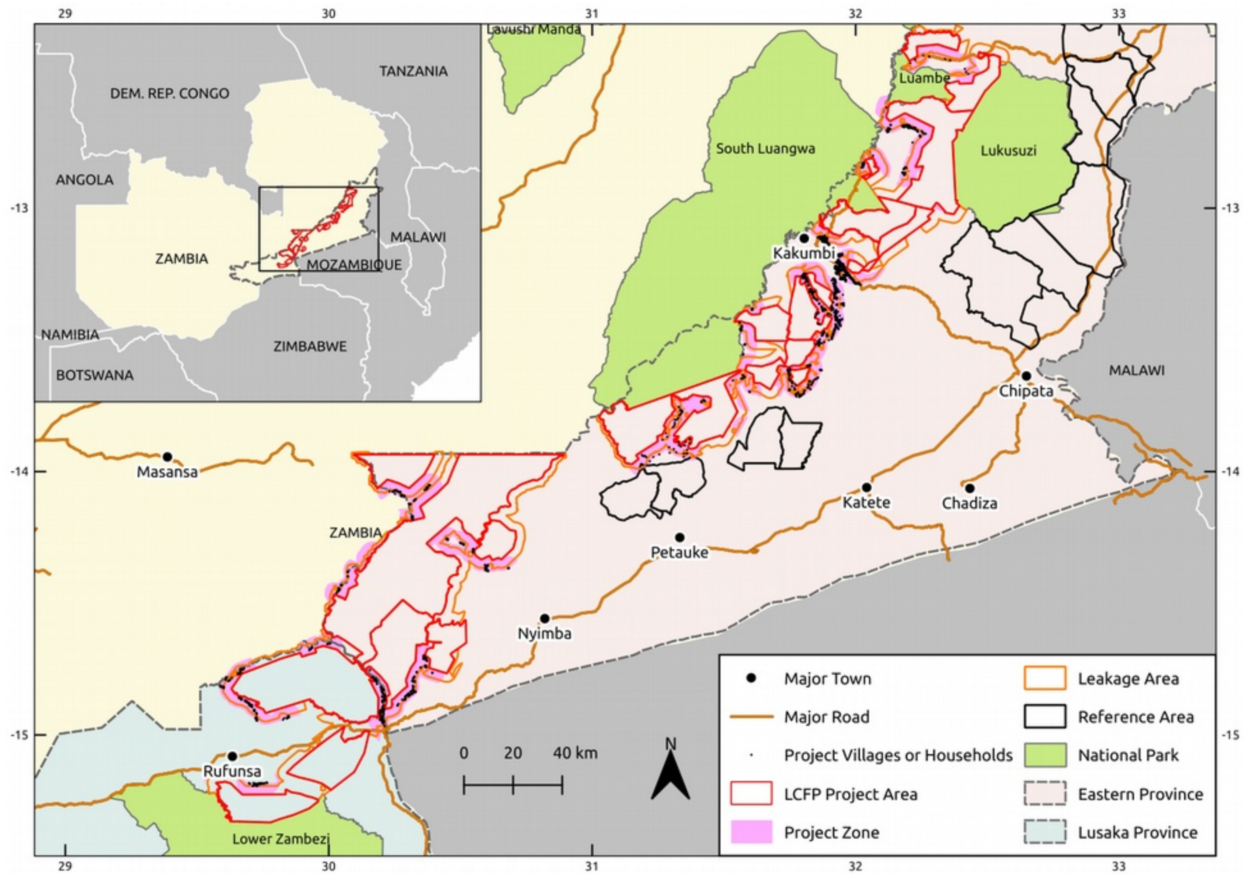

Figure 6: Project area, project zone, and leakage areas where project activities take place.

Activity Area 1: Infrastructure Development, Social Service Delivery;

Defined Activities:

- Construction and rehabilitation of school infrastructure; classroom blocks and latrines,
- Provision of safe drinking water; through drilling of new boreholes, rehabilitation of existing boreholes and improvement of existing wells.
- Provision of boats to facilitate safe river crossing
- Support the delivery of health care services through provision of clinic furniture and fittings
- Provision of hammer mills, maize and rice shellers

Additional Activities Undergoing Investigation and Further Consultation:

- Roof-based water collection (Self Supply)
- Micro-credit project(s)

- Construction and rehabilitation health care centers,
- Construction of culverts, drifts
- Erection of solar fences to reduce human – wild animal conflicts

Expected Positive Impacts:

- Reduction in cost of living
- Increased level of education in communities
- Increased local capacities
- Increased life expectancy
- Reduced child mortality and disease transmission

Potential Negative Impacts:

- Conflicts related to the equitable distribution of benefits

Activity Area 2: Apiculture, Agricultural Improvement, Diversification and Economic Opportunities

Defined Activities:

- Develop agroforestry demonstration plots to test and demonstrate agricultural multi-cropping, diversification, and intensification.
- Train 84 lead farmers in conservation farming techniques to support 1 260 follower farmers through the farmer-to-farmer model.
- Develop apiary demonstration plots, and train 6 lead farmers (mentors) in hive construction, maintenance, and appropriate apiary and beekeeping techniques.
- Distribute 6 000 beehives to 16 100 participating households within the project area.

Additional Activities Undergoing Investigation and Further Consultation:

- Support development of horticulture production
- Support animal husbandry and facilitate the construction of animal enclosures to prevent spread of diseases and damage to crops
- Assistance with commercialization of agricultural products and Support in crop aggregation for easy access to markets outside the project area
- Land-based aquaculture

Expected Positive Impacts:

- Increased quantity, diversification, and value of crops

- Improved nutritional value of crops
- More sustainable practices (Conservation farming, Beekeeping and forest management training)
- Reduced pressure on ecosystem
- Improved resilience to crop failure and market fluctuations
- Increase employment opportunities (Lead Farmers)
- Improved well-being
- Guaranteed market for honey

Potential Negative Impacts:

- Increased cost of living if value of crops increases
- Conflict related to unequal distribution of benefits and workload

Support to improved agriculture techniques through Conservation Farming

In 2017, LCFP began its conservation farming activities by supporting the work of the Ministry of Agriculture, specifically through the provision conservation farming extension services. Agreements were signed with the respective territorial Ministry of Agriculture offices within the project zone in the districts listed in Table 3.

Table 3. District agreements with Department of Agriculture.

Location	Date Agreement signed with MoA
Mambwe District	21/11/2016
Nyimba District	15/03/2017
Lundazi District	21/01/2017
Rufunsa District	29/05/2017

Arising from the FPIC community consultative meeting; forests are one of the main sources of livelihood for the local communities with illegal charcoal production, illegal logging and unsustainable farming methods such as shifting cultivation (“slash and burn”) being the major activities conducted. These activities coupled with the increased in–migration of farmers from the plateau to the valleys, where soils are still fertile, has increased the demand for opening up new farm land within the project area. Conservation Farming was thus one of the activities identified during the needs assessment process as an intervention that would complement LCFP’s conservation activities. This intervention is targeting to reach 1 260 households with conservation farming training and extension services within the first year of implementation.

For the 2017 – 2018 farming season, 2199 farmers from the target 3285 (Table 4) were enrolled by the Ministry of Agriculture to participate and receive extension services in Conservation Farming.

Table 4. Conservation farmers enrolled in project zone.

Location (District)	Total
Lundazi	375
Mambwe	1950
Nyimba	400
Rufunsa	560
Total	3,285

The main activities that are being supported under this intervention are;

- Promotion of planting of *Faidherbia albida* (Musangu) and other agro forestry species.
- Training of “facilitators” “lead” farmers in participatory research and extension techniques and improved crop production technologies.
- Strengthening of existing farmers’ groups through training in group dynamics, business administration
- Promote land management through farmer training in land preparation;

Expected Outcomes:

- Reduced encroachment into the forests under protection through REDD+
- Increased production of high value crops (soya beans, cowpeas, sunflower).
- Improvements in the income levels of rural households through the sale of surplus grains.
- Enhancement of the nutritional status of farm families, especially the children and women members through increased consumption of pulses.
- Confidence built amongst small-scale producers due to cultivating using sustainable methods and diversifying their food production, and entering into small-scale agri-business practices.
- Reduced dependence on the agriculture subsidies and use of artificial fertilisers.
- Enhancement of the knowledge base and skills of small-scale rural producers in modern production, processing and marketing practices.
- Access to the local, regional and national markets ensured small-scale rural producers through market linkages

Support to the development of Value Chains in Non- timber forest Products

Improved Apiculture techniques:

Whilst beekeeping is regarded as a cottage industry in Zambia, that is well suited for individuals and companies with limited resources for investment, its potential to contribute to the reduction of rural poverty cannot be ignored. According to the Central Statistics Office (2010), the incident of rural poverty in Zambia is about 80 %. If well organized, beekeeping can contribute to poverty reduction as it increases people's income.

Although honey hunting and beekeeping have been practiced for a long time by almost all communities in Zambia, beekeeping is only now being promoted as an alternative livelihood for those people that are involved in charcoal production and illegal game hunting, and for those that want an extra income.

Against this background, LCFP conducted a viability assessment in 2016, on the potential to establish honey production value chains within the LCFP project area. Based on the findings of this viability assessment, an agreement was signed with Beesweet, to expand honey production to cover the LCFP project zone. Through this intervention, 12 from a target of 11 honey producer- lead farmers (Mentors), received training from Beesweet in hive construction and maintenance and apiary management (Table 5).

Table 5. Number of bee mentors trained in project zone.

District	Chiefdom	No. Mentors trained	Date of Completion of training
Mambwe	Mwanya	3	July 2016
	Jumbe	2	May 2017
	Mnkhanya	1	May 2017
	Msoro	3	August 2017
	Malama	3	August 2017

Through this intervention, 6 000 Kenya topbar beehives will be distributed within the LCFP project area to 1,100 beneficiary households (Table 6).

Table 6. Beehive distribution in project zone.

District	Chiefdom	Target No, of hives to be distributed	Actual No. distributed	Date of last beehive delivery
Mambwe	Mwanya	1000	760	May 2018
Mambwe	Jumbe	1000	1000	August 2017

Mambwe	Mnkhanya	1000	1000	February 2018
Mambwe	Msoro	1000	900	May 2018
Mambwe	Malama	1000	1000	May 2018
Mambwe	Nsefu	1000	375	May 2018
Total		6000	5,035	May 2018

Expected Outcomes:

- Reduced encroachment into the forests under protection through REDD+ by local communities harvesting honey
- Increased production of natural honey.
- Improvements in the income levels of rural households through the sale of honey.
- Reduction in use of traditional hives and falling of trees due to honey hunting.
- Enhancement of the knowledge base and skills of honey producer in modern honey production, processing and marketing practices.
- Confidence built amongst honey producers due to having a guaranteed market for the honey produced.

Support to Infrastructure Development and Social Service Delivery;

Construction of School Infrastructure:

In 2016 following the community needs assessment conducted in Luembe and Nyalugwe Chiefdoms, LCFP working with community of Luembe and Nyalugwe Chiefdoms and the Ministry of Education in Nyimba constructed a 1x2 classroom block at Mkoma Primary School and one teachers house, as Partnership Impact Projects intended to benefit the community of Mkoma and Mushalira.

In Luembe Chiefdom, the construction of the teachers house commence in July of 2016 and completed in January 2017, Whilst the construction of the 1x2 classroom block at Mkoma begun in November 2016 and was completed in May 2017 for use by the community.

Construction of Clinic Infrastructure:

LCFP in Mwanya in consultation with Ministry of Health in Mwanya embarked on the construction of 1 maternity annex at Mkasanga Rural Health centre, as a Partnership Impact Project. Local skilled community members were engaged to construction the Maternity annex under the supervision and mentorship of the Buildings Officer from the Ministry of Works and Supply in Lundazi District. Construction works to the maternity annex, which started in 27/09/2016 are still on-going.

Support to delivery of Community Social Services:

As part of the community social services support through Partnership Impact Projects, in Mwanya Chiefdom, LCFP procure and supplied 1 fibreglass boat and hospital furniture to benefit the community of Mwanya Chiefdom. The boat provides safe river crossing at Lukusuzi for the community and especially the school children in Mkwela, Whilst the furniture and fittings provided was delivered to Mwanya Chiefdom for use at Yakobe Rural Health Post

In Msoro Chiefdom, LCFP provided three maize shellers, as Partnership Impact Projects intended to benefit the entire community of Msoro. LCFP handed over the maize shellers to the community of Msoro Chiefdom through the community Resources Board, who have stationed each Maize sheller in specific zone within the Chiefdom for ease of access to the beneficiary local community.

Improved water supply:

Following the community needs assessment conducted during the FPIC meetings to develop the PFMP in October 2015, LCFP drilled six new boreholes to provide clean safe drinking water to the local communities in Jumbe and Kakumbi Chiefdoms. In addition, in February 2016, LCFP commenced works to improve 10 open community wells through the construction of well lining, covers and installation of handpumps as water lifting devices (Table 7).

Table 7. Infrastructure projects in the project zone.

Location (District)	Description of Intervention	Date started	Date Completed
Nyimba	Construction of 1 teachers house	23/07/2016	24/04/2017
Nyimba	Construction of 1x2 classroom block	25/11/2016	12/05/2017
Mambwe	Construction material support towards construction of Community Guest House	15/04/2016	On - going
Lundazi	Construction of Maternity Annex	27/09/2016	On – going
Mambwe	Drilling of 6 new boreholes	12/05/2016	28/11/2016
Lundazi	Provision of Furniture and fittings	15/04/2016	10/06/2016
Lundazi	Provision of fiberglass boat	13/04/2016	18/05/2016
Mambwe	Provision of 3 maize shellers	18/07/2017	25/09/2017
Mambwe	Improving of 10 open wells	18/02/2017	On -going

Expected Positive Impacts:

- Reduction in cost of living

- Increased level of education in communities
- Increased local capacities
- Increased life expectancy
- Increased institutional deliveries and improved antenatal and Postnatal services
- Reduced child mortality and disease transmission

Potential Negative Impacts:

- Conflicts related to the equitable distribution of benefits

Our Theory of Change (Figure 9) around the provision of livelihoods support is that if we can help communities make long term investments in a shared vision for development and demonstrate that those investments are intrinsically linked to the successful long term conservation of forests, then livelihoods in those communities will improve demonstrably and be sustainable, while natural forests will remain intact.

Key risks to providing benefits to the communities include the following:

- Risk of elite capture of benefits, either from REDD+ or from livelihoods activities themselves.
- Communities will not grasp the connection between performance (REDD forest conservation) and the benefits they receive.
- Households will not benefit enough from REDD+ to halt deforestation.
- Households may “cheat” and take benefits while continuing to deforest.

To address some of these risks BCP is implementing forest and biodiversity protection measures, in addition, since these risks are complex and difficult to deal with we will also forward a Beyond Subsistence Livelihoods approach that:

- Builds the capacity of the Community Resource Boards (CRB) to manage financial resources, govern themselves effectively and build a shared vision for development.
- Provides demonstrable, regular benefits that are used to invest in the shared vision for development that benefit the household level.
- Ensures communities understand linkages between receiving REDD+ benefits and long term forest conservation.

This combination of community based natural resource governance and sustainable livelihoods development will result in long term conservation and sustainable development impacts for these communities.

Our Beyond Subsistence Livelihoods activities are focused on achieving outcomes that will set the stage for achieving development impacts. CRB’s need a vision for development implemented through effective and fair governance of the institution. Investments should lead to improved infrastructure and social services (education and health). Private sector should invest in these communities to build mutually beneficial partnerships. This approach requires that all members of the CRB “sign on” to and honor conservation covenants that spell out the responsibilities that go along with the benefits that come from REDD+.

If we can support our community partners to achieve these outcomes then these impacts will flow from our work:

- Increased income at the household level from forest conservation friendly livelihoods.
- CRB members will understand that deforestation leads to reduced benefits from REDD+.
- CRB members will be empowered to govern effectively.
- There will be no elite capture of REDD+ benefits.
- Households will benefit from improved social services.

In combination with our efforts to build local capacity and responsibility for forest and biodiversity conservation these livelihoods efforts will lead to our ultimate goals of keeping natural forest ecosystems intact and ensuring that community livelihoods are profitable and sustainable.

Activity Area 3: Forest Encroachment and Wildlife Monitoring and Poaching Reduction

BCP provides support to the local CRB village scouts and the Department of National Parks and Wildlife (DNPW) to ensure protection of the project area from encroachment in the project area, monitoring of wildlife and reducing poaching. This activity is supported by aerial monitoring.

Defined activities:

- Monthly patrols by village scouts along project area boundaries identified as encroachment hotspots.
- Monthly aerial monitoring of wildlife
- Monthly aerial monitoring of encroachment along project boundaries
- Monthly monitoring of high conservation value species
- Monthly anti-poaching activities including removal of snares, counting carcasses, apprehending poachers.

Expected Outcomes:

- Quick response to encroachment in project area
- Rapid resolution of encroachment through consultation process by local governance structures notably the CRB and chiefs.
- Reduction of encroachment over time as project activities expand and awareness of monitoring activities take hold

- Greater understanding of wildlife populations and threats to wildlife.
- Adapt management to changing wildlife populations.
- Increase in wildlife populations, especially high conservation value species.
- Employment creation for village scouts.
- Reduction of leakage and project emissions.

Thus far monitoring has been rolled out in most of the project area by participating CRBs. BCP has developed a SOP for monitoring which outlines the interaction between aerial and ground based monitoring. Most encroachment is detected through aerial monitoring and recorded using a GPS. This information is then stored in a database and follow up ground monitoring is then actioned by BCP conservation staff. Scouts are given the location of encroachment, once confirmed on the ground and contact made with encroachers, the Community Engagement Team and CRB then follow up and investigate incidences further.

Monitoring has been rolled out in most of the project area and results of aerial monitoring can be seen in Figure 7 and ground monitoring in Figure 8.

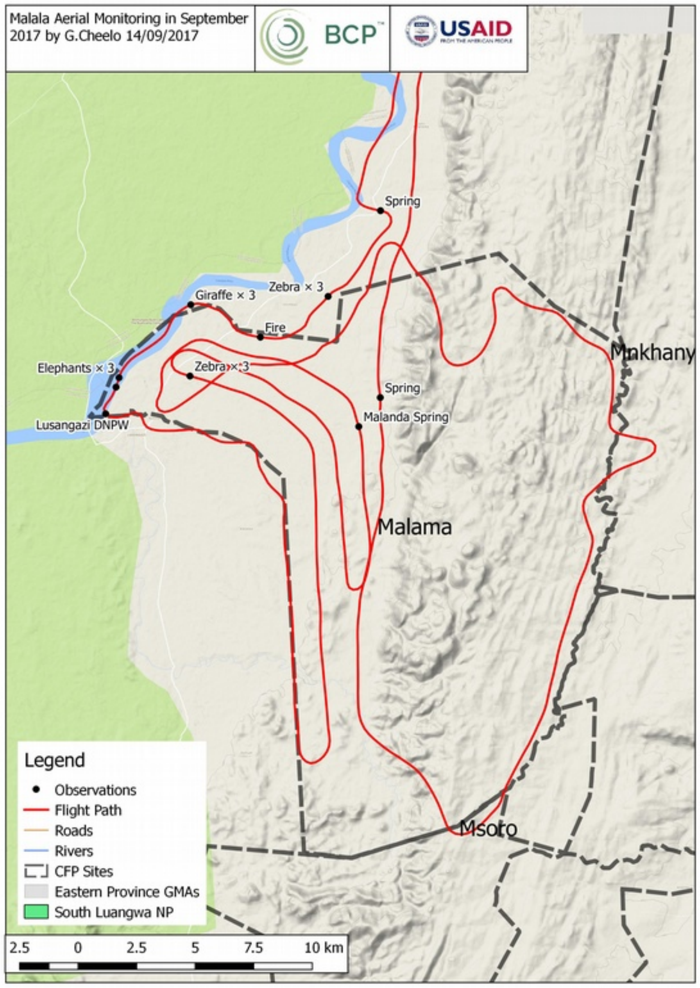

Figure 7. Aerial monitoring map of project area.

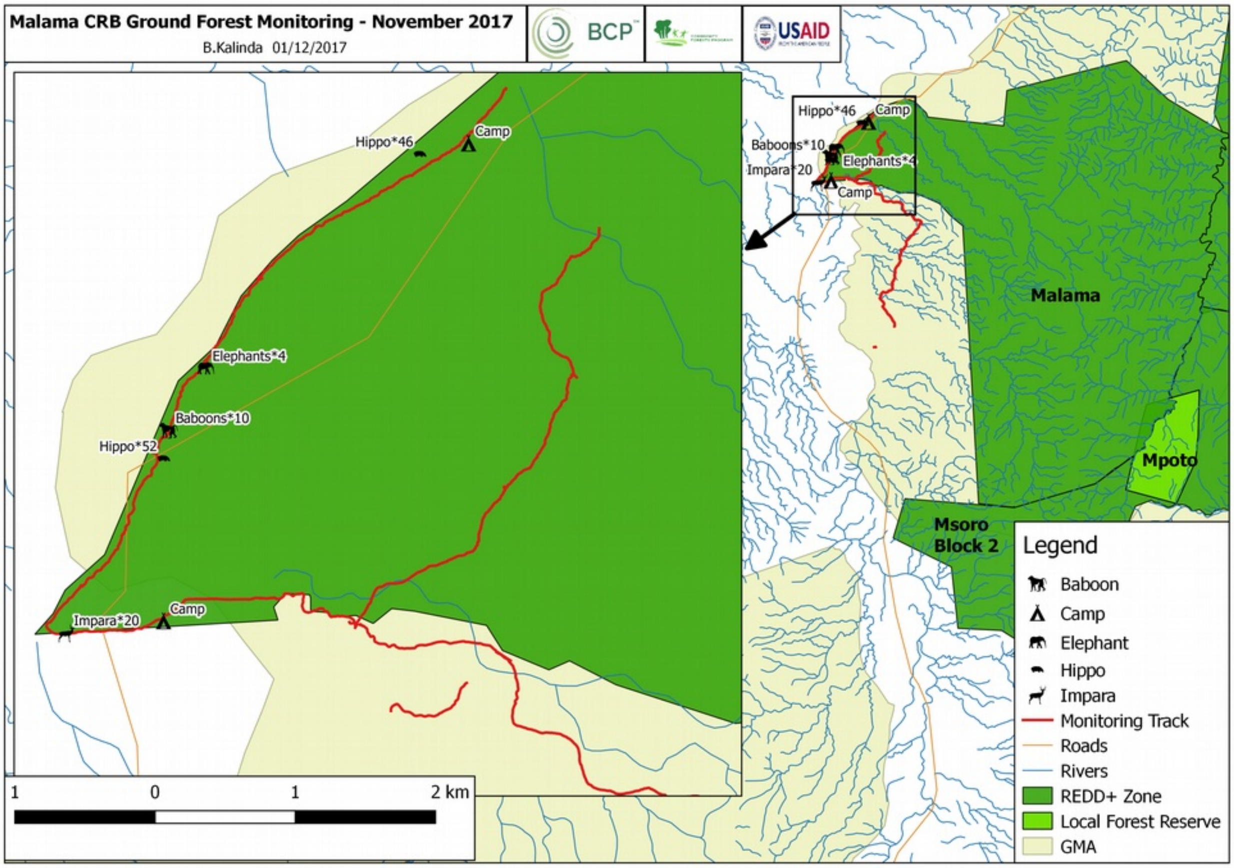

Figure 8. Ground monitoring of project area.

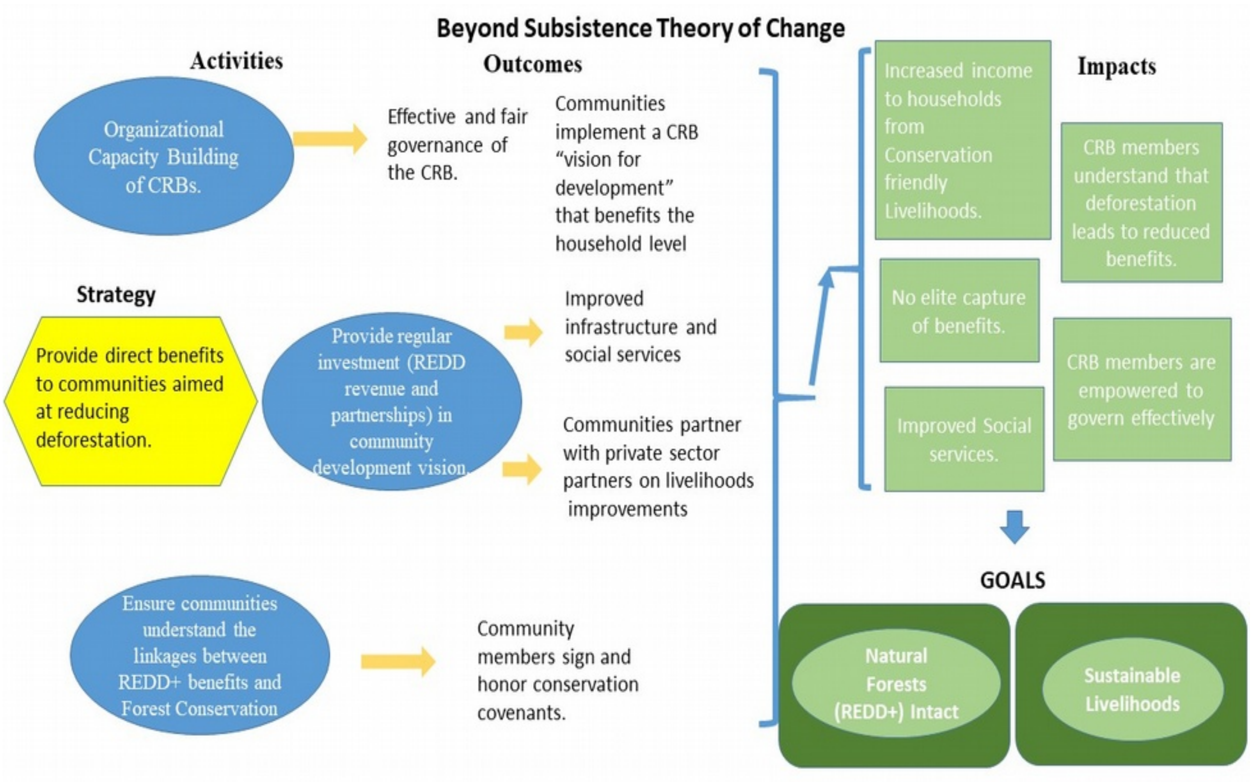

Figure 9. BCP Community Theory of Change

All of the above mentioned activities are connected in some way to reducing the non-permanence risk by ensure as little deforestation takes place in the project area, community members share are benefited by the project, hence reducing opportunity cost risks and activity displacement leakage is limited to a minimum.

2.2 Deviations

Methodology Deviations

MR.97

Demonstration of Forest Age

The methodology requires (in PDR.22) that a digital GIS-based) map of the project accounting areas, including aerial or satellite imagery showing that they are comprised of forests or native grasslands as of the project start date and 10 years prior to the project start date. The project did not meet this requirement because it chose to map forests in the project area with PALSAR radar data and the earliest available data for that sensor is from 2007. However, this is justified on several grounds. The project areas are virgin forests that were previously remote and are inaccessible. Thus, there is no history of large disturbances in the area that would have been followed by regeneration between 2005 and 2007. An accurate 2007 forest map of these areas should be virtually indistinguishable from a 2005 map.

Additionally, using the 2007 PALSAR imagery results in a much more accurate map of the forests that existed 10 years prior to the start date than would be possible with optical imagery. The optical imagery best suited to mapping the project area is generally from April-June. It is at this time of year, that relatively cloud free imagery can be obtained when the forests still have leaves and are not dotted with burn scars. Even so, the passing of just a couple of weeks during this period, which is immediately after the rains, results in significant changes in the appearance of the forest in optical imagery. The project initially tried mapping forests with optical imagery, but found significant changes between years that were the result of seasonality rather than real changes in forest areas. These effects are much reduced when using PALSAR radar data. This is because, in addition to being cloud penetrating, L band radar has a relatively long band-width and thus does not generally interact with the leaves of trees, but does interact with large branches and stems[2]. Finally, VM0009 does not require a remote sensing analysis of an area, but rather to visual show that it is unconverted as of the project start date and 10 years prior. Such an approach is highly subjective since unconverted grasslands and farmlands may at times appear similar. With the PALSAR based analysis, all non-forest areas, whether natural or not were excluded. Thus, using the PALSAR data improved the accuracy of the analysis and did not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. The deviation only affects the monitoring and measuring part of the methodology.

Leakage Plot Protocol

There was a minor deviation of the visual approach for leakage monitoring (Appendix R, VM0009). Instead of Forest Inventory Team members estimating degradation, they were required to geo-reference every felled tree in the permanent plots and the total number of felled trees were compared against the average density of trees/ha in the project accounting area, again avoiding biased estimations/erroneous counting of trees in 1ha plots. This value was then placed in a degradation category as per VM0009. This approach removed all potential bias and greatly improves the robustness and repeatability of degradation monitoring. This deviation does not affect the conservativeness of any emissions reductions as it is in fact an improvement of the prescribed procedure and removes observer bias that could result in degradation in the leakage area being underestimated. The deviation only affects the monitoring and measuring part of the methodology.

Biomass estimation

The project plans to use two different allometric equation to estimate above ground biomass. One was developed from data provided by Prof. Chudimayo from Zambian Miombo (see Appendix G) and was tested on trees up to 55 cm DBH. It will be used to estimate the biomass of trees up 63 cm dbh, with the biomass of trees between 55 cm and 63 cm dbh assumed to be equal to a 55 cm dbh tree. Then, a second equation developed by Mugasha et al. in Tanzanian Miombo (see Appendix G) will be used to estimate the biomass of trees between 63 cm dbh and 110 cm dbh with trees with a greater than 110 cm dbh assumed to have a biomass equal to a 110 cm dbh tree. This parsing is necessary because the methodology requires that the directly measured biomass of the largest tree in the representative sample of destructively harvested trees be greater than the biomass predicted by the equation, and that the value of the first order derivative of the largest tree in the representative sample and inventory not differ by more

2 Mitchard, E. T. A., et al. "Using satellite radar backscatter to predict above □ ground woody biomass: A consistent relationship across four different African landscapes." Geophysical Research Letters 36.23 (2009)

than 10%. The first selected allometric equation does under predict the biomass of the largest tree in the representative sample, but the difference of the first order derivative values was greater than 10%. The second allometric equation doesn't past the first test (it slightly over predicts the biomass of the largest tree in the sample). However, compared to the Zambian equation, it predicts lower biomass for every size tree. Therefore, the equations were applied as described above to enable the project to make estimate of the biomass of larger trees that fell outside the range of the size of trees used to develop the equations. Only 33 trees out of 3589 trees in the inventory had a dbh above 55 cm and only 3 had a dbh above 110 cm. Applying the equations in this way is conservative and erases the possibility of the equation over predicting biomass of trees outside the size classes for which it was tested. This deviation only relates to how biomass is measured for monitoring purposes and doesn't affect any other part of the methodology. It results in a more conservative estimate of biomass and emissions savings. Therefore, it meets the criteria for an acceptable methodological deviation.

Project Description Deviations

As stated in the project description, the project plans to consider the SOC carbon pool in the calculation of emissions. However, SOC accounting area data was not yet available at verification and thus the project has decided to conservatively exclude SOC from this monitoring period. The deviation is conservative as SOC is expected to be lower in the baseline scenario than in the project scenario. The deviation only relates to a change in monitoring and does not impact any other aspect of the methodology and is therefore an acceptable methodological deviation.

2.3 Grouped Project

There are no new project instances for this monitoring period.

MR.7

The project activity start date for initial project area instance is the same as the project start date: May 1st, 2019.

MR.8

Therefore, t_{PAI} (the number of days since the project start date) for the initial project area instance is 0.

2.4 Safeguards

No Net Harm

The project is seeking validation under the CCB Standard Version 3 and any potential negative impacts are summarized in the joint VCS/CCB Project Description. The project is an avoided unplanned deforestation project and specifically to conserve the astounding biodiversity of the project area and benefit small holder farmers dependent on the project area with a variety of project activities aimed at improving livelihoods through poverty reduction such as conservation agriculture and beekeeping.

Local Stakeholder Consultation

Stakeholder identification took place as part of the initial REDD+ viability assessment process that BCP undertook to identify new / potential REDD+ sites for the LCFP, beginning in 2014. The process followed can be broadly outlined as follows:

Workshops / Meetings with key Government and Community Authorities: Beginning at a national level, and then moving down to provincial, district and Chiefdom levels, BCP held a series of workshops and meetings, in which we outlined REDD+ and explained the REDD+ viability assessment process, including two key criteria for REDD+:

- Selection of a viable forest area for protection;
- Documented consent from all relevant stakeholders / rights-holders.

Permissions were then received from Government and Chiefdom authorities, to proceed with on-the-ground viability assessment activities, including participatory forest identification and stakeholder engagement in line with FPIC.

Initial "high level" stakeholder identification and mapping: During the initial District and Chiefdom-level workshops and meetings that were held, BCP solicited inputs from relevant Government and community authorities (namely: traditional authorities, including Chiefs / Chiefs' representatives, and CRBs, as legally recognized, Chiefdom-level community institutions), about who could be counted as potential stakeholders to any proposed REDD+ project, and where they were/are located, in relation to the proposed forest areas under consideration for REDD+ protection. It was explained that identifying and consulting with any potential stakeholders is a verification requirement for REDD+, and more importantly, adequate stakeholder sensitization and consultation prior to project implementation is in everyone's interest, to help ensure the sustainability of the project, by ensuring that all relevant stakeholders were consulted about, and consented to, participation in the project. It was explained that stakeholders could broadly be defined as anyone who could potentially be impacted by the proposed project (either positively or negatively), and/or anyone with potential decision-making power about whether or not to cut trees within the identified / proposed forest areas for protection.

At the District/Chiefdom level, Government and community authorities (traditional leaders) provided us with consistent feedback that, so long as a REDD+ project would be taking place on “community” land (including within a GMA), then all community members of the Chiefdom could be considered to be stakeholders, since traditional authorities recognize that community forests belong to the entire community within the Chiefdom, and therefore, all recognized residents within the Chiefdom would have recognized rights, within the traditional leadership structure, to access and use forest resources within community forest areas. Initial community maps were drawn by these key stakeholders, and also confirmed and supplemented by satellite imagery, which confirmed the location of villages and settlements, near to proposed forest areas within each Chiefdom that had expressed willingness / interest to participate in the program.

Detailed Chiefdom/Village level stakeholder mapping - PRA/PLA FPIC activities: More detailed stakeholder identification and mapping took place during significant PRA/PLA techniques that were undertaken in each of the 12 partner Chiefdoms, as part of the FPIC processes that were undertaken by BCP during 2015-2017, at a community level. Our FPIC records show that these meetings were held at a Chiefdom level, in each of the 12 Chiefdoms, and that representatives from all 62 registered VAGs within these 12 Chiefdoms were involved in FPIC activities. During the PRA/PLA activities that took place, community stakeholder identifications were confirmed, and maps were further refined, to ensure that significant community inputs were received to identify and include any potential stakeholder communities. Local (District) level Government officers were involved in this process, as well as representatives from Chiefdom level institutions (Chiefdom royal establishments; CRBs), as well as village-level representatives from VAGs and traditional leaders (Headmen). Meetings were also open to members of the wider community, who also participated and provided inputs. As part of the stakeholder identification process, it was described, again, that “stakeholders” included anyone who could potentially be affected by the project, and/or anyone with a recognized “right,” or even ability, to influence decision-making about whether or not to cut trees within the identified forest area for protection. Again, stakeholders at all levels confirmed that community forests were/are viewed to be collectively “owned” community resources, and as such, “stakeholders” should include all Chiefdom inhabitants, in all villages within participating Chiefdoms, as their rights to access and use forest resources were/are recognized by members of the local community, as well as the Chiefdom authorities (traditional leaders) as well as Government – who recognize the authority of community institutions and traditional leaders, on community or “customary” land, even within GMAs. As such, it was agreed that a registry of Chiefdom inhabitants and villages could essentially function as a stakeholder registry. Initial lists of villages were received from Headmen, and were shared with BCP through VAG and CRB structures. These were used to inform data-collection plans, to complete an on-the-ground confirmation activity, to confirm, register and map all stakeholders for REDD+ projects within each of the 12 partner Chiefdoms of the LCFP.

Once FPIC processes were completed, community consent was received and documented in community consent letters for REDD+ implementation, signed by Chiefdom authorities with legally recognized authority to manage community resources (Chiefs, CRBs). Once these letters were received, BCP undertook a “Stakeholder Household Registry” process, with the aim of confirming, documenting, registering, mapping and counting community stakeholders to each REDD+ project, in every participating Chiefdom, in every VAG and village within that Chiefdom. This process was undertaken through CRBs and Village Action Groups (VAGs) as local-level community institutions, and involved significant inputs, guidance and participation of local traditional leaders (such as Headmen) to ensure that all members of their villages were registered as participants, stakeholders and beneficiaries to the project.

The data collection process was led by locally hired staff, working hand-in-hand with local traditional leaders such as village headmen, as well as representatives from local VAGs, CRBs, and Government.

The LCFP "Household Registry" now provides a comprehensive database, and map, of all stakeholders to REDD+ projects supported under the LCFP. As the processes above describe, it has been informed by significant inputs from community authorities and government partners, solicited through a multitude of workshops, meetings, and PRA/PLA activities that have been undertaken as part of FPIC processes during the past three years, during which input was solicited, and received, concerning who should be counted as "stakeholders," based on their recognized rights and abilities to inform decisions about forest protection within identified areas. The actual data collection process has taken approximately one year, and has involved dozens of data collectors across all 12 Chiefdoms.

Other stakeholders are hunting concession holders that operate within the REDD+ project areas.

Project documentation has been made available to stakeholders throughout the project development process, including Free, Prior and Informed Consent (FPIC) stakeholder sensitization activities that were initiated in 2014, and which have continued through 2017. Informational handouts clearly outlining the proposed project model, and accurately describing REDD+ activities, were developed and have been widely distributed to community (and Government) implementing partners, throughout this time. These handouts, and other informational materials (such as informational posters about REDD+, BCP and the CFP) have been distributed to each of the 12 partner Community Resources Boards (CRBs) that BCP is working with, to implement REDD+ activities under the LCFP. Throughout nearly 3 years, from 2015-2017, BCP has provided capacity building training to CRB members, as well as local Government implementing partners, and locally hired staff, to ensure their ability to accurately explain and disseminate information about REDD+, in line with FPIC requirements. BCP staff also developed and launched a standard "sensitization program," that provided critical information about REDD+ and the proposed activities of the LCFP, and which made information available in a multitude of local languages, presented by locally hired and trained staff, who ensured that this information was disseminated in locally appropriate ways. In combination with these sensitization programs, BCP has ensured that informational materials have been made available to local community stakeholders in each Chiefdom, and that there are trained staff and implementing partners available at all times, in each Chiefdom, to undertake ongoing sensitization activities, to accurately disseminate information about REDD+, and to be available to answer any questions or concerns that stakeholders may have, at any given time.

BCP has detailed records of all community and stakeholder meetings that will be made available to the VVB. Further details of stakeholder consultation can be found in the LCFP Project Description.

3 DATA AND PARAMETERS

3.1 Data and Parameters Available at Validation

Data / Parameter	α
Data unit	Unitless
Description	Combined effects of β and θ at the start of the historic reference period
Equations	[F.3], [F.5], [F.6], [F.8]
Source of data	Reference area and historic reference period
Value applied	0.2188
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	β
Data unit	Unitless
Description	Effect of time on the cumulative proportion of conversion over time
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.8]
Source of data	Reference area and historic reference period
Value applied	0.0003271
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	γ
Data unit	Days
Description	Time shift from beginning of historic reference period to project start date
Equations	[F.3], [F.6]
Source of data	Historic reference period
Value applied	10,589
Justification of choice of data or description of	Time in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document.

measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	Provide any additional comments

Data / Parameter	λ_{SOC}
Data unit	Proportion (unitless)
Description	Exponential soil carbon decay parameter
Equations	[F.9], [F.33]
Source of data	VCS AFOLU default for tropical regions
Value applied	0.2
Justification of choice of data or description of measurement methods and procedures applied	Conservative default based on IPCC value.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	$\hat{\sigma}_{EM}$
Data unit	Standard deviation (unitless)
Description	The estimated standard deviation of the state observations used to fit the logistic function
Equations	[F.12], [F.14], [B.31]
Source of data	Remote sensing image interpretation. Analysis undertaken as described in section 3.2.1 of the project document.
Value applied	0.4273
Justification of choice of data or description of measurement methods and procedures applied	Time and place in which the logistic model is fit. Analysis undertaken as described in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	B
Data unit	Set
Description	The set of all selected carbon pools in biomass. Is a subset of *C*
Equations	[F.17], [F.18], [F.23], [F.50]
Source of data	PDD

Value applied	2
Justification of choice of data or description of measurement methods and procedures applied	AGOT and BGOT, See section 1.1.1 of the project document for justification
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	C
Data unit	Set
Description	The set of all selected carbon pools
Equations	
Source of data	Monitoring records
Value applied	3
Justification of choice of data or description of measurement methods and procedures applied	AGOT, BGOT, SOC, See section 1.1.1 of the project document for justification
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	J
Data unit	Set
Description	The set of all observations of conversion. When superscripted with a monitoring period, the conversion observations are taken for leakage analysis
Equations	[F.13]
Source of data	Remote sensing image interpretation or field observations in the leakage area
Value applied	13,194
Justification of choice of data or description of measurement methods and procedures applied	Analysis undertaken as described in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	M
Data unit	Set

Description	The set of all monitoring periods
Equations	[F.32], [F.33], [F.36], [F.54], [F.56]
Source of data	Monitoring records
Value applied	27
Justification of choice of data or description of measurement methods and procedures applied	The project crediting period is 30 years and after this first verification, the project will use annual verification. Thus there will be 27 verification.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	A_{PAA}
Data unit	ha
Description	Area of project accounting area
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.8], [F.41], [F.52], [F.57]
Source of data	GIS analysis prior to sampling
Value applied	485,495
Justification of choice of data or description of measurement methods and procedures applied	Area within the project area that is on slopes of less than 45%, less than 6.5 km from a road, and was forest at least 10 years before the project start date. Described in more detail in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	A_{PX}
Data unit	ha
Description	Area of proxy area
Equations	[F.57]
Source of data	GIS analysis prior to sampling
Value applied	3840
Justification of choice of data or description of measurement methods and procedures applied	Areas converted from forest to non-forest between 2010 and 2017 falling within 2 km of buffer around roads passing through communities favorable to having proxy plots on their lands. Described in more detail in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	n_d

Data unit	
Description	Number of spatial points in the reference area
Equations	[F.14]
Source of data	Remote sensing image interpretation
Value applied	2600
Justification of choice of data or description of measurement methods and procedures applied	Value deemed sufficient as per VM0009 to adequately quantify forest state in the reference area. Described in more detail in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	o_i
Data unit	Binary
Description	State observation for the sample point in the reference area
Equations	[F.13]
Source of data	Remote sensing image interpretation
Value applied	0 or 1
Justification of choice of data or description of measurement methods and procedures applied	Data collected as per the methodology. Described in more detail in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	P_{LME}
Data unit	Proportion (unitless)
Description	Portion of leakage related to market
Equations	[F.51]
Source of data	Analysis of drivers of deforestation (Appendix K)
Value applied	0
Justification of choice of data or description of measurement methods and procedures applied	Few wood products are taken from the project area in the baseline scenario and most farming in the baseline scenario is subsistence.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	q
Data unit	Days
Description	Lag between start of degradation and conversion
Equations	[F.3], [F.4], [F.5]
Source of data	Consultation with agents of deforestation in near project area in Lusaka province – 28 farmers were questioned to determine how long they usually wait after clearing an area before ploughing it.
Value applied	620
Justification of choice of data or description of measurement methods and procedures applied	Based on observations in the reference area, hence applicable to the project area.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	r_{RS}
Data unit	Unitless
Description	Expansion factor for above-ground biomass to below-ground biomass (root/shoot ratio)
Equations	[F.30]
Source of data	Chidumayo, E. N. 2013. Forest degradation and recovery in a miombo woodland landscape in Zambia: 22 years of observations on permanent sample plots. Forest Ecology and Management 291:154-161.
Value applied	0.54
Justification of choice of data or description of measurement methods and procedures applied	The root-shoot ratio was determined within a study site that overlaps one of the same provinces the project is in.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	r_{U}
Data unit	Proportion (unitless)
Description	Onset proportion of conversion immediately adjacent to project area
Equations	[F.4]
Source of data	GIS analysis and image interpretation
Value applied	0.084

Justification of choice of data or description of measurement methods and procedures applied	Result of remote sensing analysis described in more detail in section 3.2.1 of the project document.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	t
Data unit	Days
Description	Time since project start date
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.8], [F.10]
Source of data	Calendar
Value applied	1340
Justification of choice of data or description of measurement methods and procedures applied	Project start date is May 1st, 2015 and first monitoring period ended December 31st, 2018.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	t_i
Data unit	Days
Description	The point in time of the observation made at point i
Equations	[F.11], [A.6]
Source of data	Remote sensing image interpretation
Value applied	1-11341
Justification of choice of data or description of measurement methods and procedures applied	The historical reference period stretches from May 4th, 1986 to May 22nd 2017, which is 11341 days in total.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	t_{PL}
Data unit	Days
Description	Length of project crediting period
Equations	[F.5]
Source of data	PD

Value applied	10957
Justification of choice of data or description of measurement methods and procedures applied	The project crediting period is from May 1st, 2015 to April 30th, 2045 which is equivalent to 10957 days.
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	w_i
Data unit	Unitless
Description	Weight applied to the i th sample point in the reference area
Equations	[A.6], [F.13]
Source of data	Remote sensing image interpretation
Value applied	Varies (see Appendix O)
Justification of choice of data or description of measurement methods and procedures applied	Calculated as specified in VM0009
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	$x_i\, y_i$
Data unit	Geographic coordinates
Description	Latitude and Longitude of the i th sample point
Equations	[A.6], [F.11]
Source of data	Remote sensing image interpretation
Value applied	Varies (see appendix Appendix O)
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	Calculation of baseline emissions
Comments	

3.2 Data and Parameters Monitored

MR.85 - 87

Data / Parameter	$A_{P1}^{[m=0]}$
Data unit	ha
Description	Area of project accounting area stratum 1 prior to first verification event
Equations	[F.24]
Source of data	GIS analysis prior to sampling
Description of measurement methods and procedures to be applied	GIS analysis removing areas in the project area that were non-forest prior to the project start date, on slopes greater than 45% or in areas farther than 6.5 km from a road.
Frequency of monitoring/recording	First monitoring period / 10 years
Value applied	485,495
Monitoring equipment	N/A
QA/QC procedures to be applied	Double checking GIS analysis
Purpose of data	Calculation of baseline emissions
Calculation method	GIS area calculation
Comments	

Data / Parameter	$c_B^{[m]}$
Data unit	tCO_2e / ha
Description	Baseline carbon stocks at the end of the current monitoring period
Equations	[F.2], [F.3], [F.4], [F.5], [F.6], [F.7], [F.57]
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol Appendix P.
Frequency of monitoring/recording	20% of plots measured annually
Value applied	15.6
Monitoring equipment	See Appendix P. Standard vegetation quantification equipment.
QA/QC procedures to be applied	See Appendix P and description in monitoring report.
Purpose of data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations.

Comments	

Data / Parameter	$C_{B\ BGB}^{[m]}$
Data unit	tCO_2e
Description	Carbon not decayed in BGB at the end of the current monitoring period
Equations	[F.16]
Source of data	Equation F.32 from VM0009
Description of measurement methods and procedures to be applied	Ongoing forest monitoring as per Appendix P
Frequency of monitoring/recording	Based on peer reviewed literature and prescribed equation in VM0009
Value applied	83,745
Monitoring equipment	See Appendix P
QA/QC procedures to be applied	See Appendix P
Purpose of data	Calculation of baseline emissions
Calculation method	Applied equation F.32 in VM0009 3.0
Comments	

Data / Parameter	$C_{B\ SOC}^{[m]}$
Data unit	tCO_2e
Description	Carbon not decayed in SOC at the end of the current monitoring period
Equations	[F.16]
Source of data	Project accounting area sampling and application of equation F.33 from VM0009 3.0
Description of measurement methods and procedures to be applied	Accounting area sampling as per the protocol in Appendix P and application of equation F.33, VM0009 V3.0
Frequency of monitoring/recording	Fixed once off sampling, not repeated as per VM0009
Value applied	N/A SOC excluded for this monitoring period (see section 2.2)
Monitoring equipment	See Appendix P standard soil sampling procedures and equipment.
QA/QC procedures to be applied	See Appendix P.
Purpose of data	Calculation of baseline emissions
Calculation method	Application of equation F.33 VM0009 V3.0 using SOC

	values from the project accounting area.
Comments	SOC data at validation comes from LZRP REDD project as LCFP SOC data not ready at document submission date.

Data / Parameter	$c_{B\,b}^{[m]}$
Data unit	tCO_2e / ha
Description	Baseline scenario average carbon stock in selected carbon pools
Equations	[F.18] [F.24]
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol Appendix P.
Frequency of monitoring/recording	20% of plots measured annually
Value applied	15.6
Monitoring equipment	See Appendix P. Standard vegetation quantification equipment.
QA/QC procedures to be applied	See Appendix P and description in monitoring report.
Purpose of data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations.
Comments	

Data / Parameter	$c_{B\,BM}^{[m]}$
Data unit	tCO_2e / ha
Description	Baseline carbon stocks in biomass at the end of the current monitoring period
Equations	[F.19], [F.20], [F.21], [F.24], [F.52]
Source of data	Proxy area sampling
Description of measurement methods and procedures to be applied	Proxy area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	At first verification and baseline revision
Value applied	15.6
Monitoring equipment	See Appendix P
QA/QC procedures to be applied	Review of monitoring records as described in Appendix P and re-sampling of 20% of plots

Purpose of data	Calculation of baseline emissions
Calculation method	Using standard vegetation ecology plot sampling techniques and subsequent quantification of biomass using allometric equations. Averages calculated as per equation B.9 VM0009 V3.0
Comments	

Data / Parameter	$c_{B\ SOC}^{[m]}$
Data unit	tCO_2e / ha
Description	Baseline soil carbon stocks at the end of the current monitoring period
Equations	[F.25], [F.27], [F.28]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	Once at first verification
Value applied	N/A. SOC is excluded in this monitoring period (see section 2.2)
Monitoring equipment	See Appendix P, standard soil sampling equipment
QA/QC procedures to be applied	Review of monitoring records at next baseline re-evaluation
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of SOC done according to Annex B.2.6, VM0009 3.0
Comments	

Data / Parameter	$c_{P}^{[m]}$
Data unit	tCO_2e / ha
Description	Project carbon stocks at the end of the current monitoring period
Equations	[F.41], [F.57]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Plot monitoring as per the protocol in Appendix P
Frequency of monitoring/recording	20% of plots monitored annually
Value applied	N/A. SOC is excluded in this monitoring period (see section 2.2)

Monitoring equipment	See Appendix P, standard vegetation sampling equipment
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_P^{[m-1]}$
Data unit	tCO_2e / ha
Description	Project carbon stocks at the beginning of the current monitoring period
Equations	[F.41]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of plots monitored annually
Value applied	236.2
Monitoring equipment	Review of monitoring records, see Appendix G.
QA/QC procedures to be applied	Review of monitoring records, re-sampling of 20% of plots.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_P^{[m=0]}$
Data unit	tCO_2e / ha
Description	Project carbon stocks prior to first verification event
Equations	[F.7]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period

Value applied	236.2
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_{P\ 1\ BM}^{[m=0]}$
Data unit	tCO_2e / ha
Description	Project carbon stocks in stratum 1 prior to first verification event
Equations	[F.24]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	236.2
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_{P\ BM}^{[m=0]}$
Data unit	tCO_2e
Description	Project carbon stocks in biomass prior to first verification event
Equations	[F.19], [F.20], [F.21], [F.52]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P

Frequency of monitoring/recording	First monitoring period
Value applied	114,668,494
Monitoring equipment	Appendix P describes the plot based monitoring procedure.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_{P\,b}^{[m]}$
Data unit	tCO_2e / ha
Description	Average carbon in biomass in the project accounting area
Equations	[F.17]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	20% of biomass plots annually
Value applied	236.2
Monitoring equipment	Appendix P describes the plot based monitoring equipment.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_{P\,1\,b}^{[m]}$
Data unit	tCO_2e / ha
Description	Average carbon in biomass for project accounting area stratum 1
Equations	[F.23]
Source of data	Project accounting area sampling
Description of measurement	Project accounting area sampling as per the protocol in

methods and procedures to be applied	
Frequency of monitoring/recording	20% of biomass plots annually
Value applied	236.2
Monitoring equipment	Appendix P describes the plot based monitoring equipment.
QA/QC procedures to be applied	Review of monitoring records, see Appendix G.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard plot sampling techniques to quantify vegetation and application of allometric equations. Averages calculated using equation B.9 VM0009 V3.0.
Comments	

Data / Parameter	$c_{P\,SOC}^{[m=0]}$
Data unit	tCO_2e / ha
Description	Project soil carbon stocks prior to first verification event
Equations	[F.25], [F.27], [F.28]
Source of data	Project accounting area sampling
Description of measurement methods and procedures to be applied	Project accounting area sampling as per the protocol in Appendix P
Frequency of monitoring/recording	First monitoring period
Value applied	N/A. SOC is excluded in this monitoring period (see section 2.2)
Monitoring equipment	Appendix P, standard soil sampling equipment
QA/QC procedures to be applied	Quantification of SOC done according to Annex B.2.6, VM0009 3.0 at soil lab.
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of SOC done according to Annex B.2.6, VM0009 3.0 at soil lab.
Comments	This soil data is borrowed from LZRP project since LCFP accounting area soil data was not yet available at submission.

Data / Parameter	$E_{\Delta GER}^{[m]}$
Data unit	tCO_2e
Description	GERs for the current monitoring period
Equations	[F.55], [F.57]

Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Every monitoring period
Value applied	3,973,861
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of GER calculations
Purpose of data	Calculation of project emissions
Calculation method	Calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0
Comments	

Data / Parameter	$E_{\Delta GER}^{[i]}$
Data unit	tCO_2e
Description	GERs for monitoring period i
Equations	[F.54]
Source of data	Area measurements
Description of measurement methods and procedures to be applied	Accounting area plots monitoring as per Appendix P.
Frequency of monitoring/recording	Prior monitoring period
Value applied	92,447,334
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of GER calculations
Purpose of data	Calculation of project emissions
Calculation method	Calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0
Comments	This estimates assumes SOC emissions factors will be similar to BCP's other REDD+ project once SOC field data is finalized for this project.

Data / Parameter	$E_{\Delta NER}^{[i]}$
Data unit	tCO_2e
Description	NERs for monitoring period i
Equations	[F.56]
Source of data	Area measurements
Description of measurement	Accounting area plots monitoring as per Appendix P.

methods and procedures to be applied	
Frequency of monitoring/recording	Prior monitoring period
Value applied	81,864,979
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Review of NER calculations
Purpose of data	Calculation of project emissions
Calculation method	Calculated with VCS AFOLU Non-Permanence Risk Tool and equation F.55 as per section 8.4.3 of VM0009 V3.0
Comments	This estimates assumes SOC emissions factors will be similar to BCP's other REDD+ project once SOC field data is finalized for this project.

Data / Parameter	$E_B^{[m]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions at the end of the current monitoring period
Equations	[F.15]
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.16.
Frequency of monitoring/recording	Every monitoring period
Value applied	4,438,058
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Equation F.16 from VM0009 V3.0
Comments	

Data / Parameter	$E_B^{[m-1]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions at the beginning of the current monitoring period
Equations	[F.15]
Source of data	Proxy area measurements
Description of measurement	Proxy area plots monitoring as per Appendix P and

methods and procedures to be applied	equation F.16.
Frequency of monitoring/recording	Prior monitoring period
Value applied	0
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Equation F.16 from VM0009 V3.0
Comments	This is the first monitoring period

Data / Parameter	$E_{B\Delta}^{[m]}$
Data unit	tCO_2e
Description	Change in baseline emissions
Equations	[F.9], [F.10], [F.14], [F.53], [F.57]
Source of data	Proxy area measurements
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P and equation F.15.
Frequency of monitoring/recording	Every monitoring period
Value applied	4,438,058
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Equation F.15 from VM0009 V3.0
Comments	

Data / Parameter	$E_{B\Delta BGB}^{[i]}$
Data unit	tCO_2e
Description	Change in baseline emissions from below-ground biomass during monitoring period i
Equations	[F.32]
Source of data	Monitoring the proxy area
Description of measurement methods and procedures to be applied	Proxy area plots monitoring as per Appendix P
Frequency of monitoring/recording	Every monitoring period

Value applied	28,151,222
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root:shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{B\Delta SOC}^{[m]}$
Data unit	tCO_2e
Description	Baseline change in emissions from soil carbon
Equations	[F.15],[F.33]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Soil sampling in project accounting area as per Appendix P
Frequency of monitoring/recording	At first verification
Value applied	N/A. SOC data is still being gathered (see section 2.2)
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of SOC using standard soil sampling techniques and application of equation F.7 and F.29 of VM0009 V3.0, the SOC decay emissions model
Comments	

Data / Parameter	$E_{B\Delta SOC}^{[i]}$
Data unit	tCO_2e
Description	Baseline emissions from soil carbon in monitoring period i
Equations	[F.33]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Soil sampling in project accounting area as per Appendix P

Frequency of monitoring/recording	At first verification
Value applied	N/A. SOC data is still being collected (see section 2.2)
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of SOC using standard soil sampling techniques and application of equation F.7 and F.29 of VM0009 V3.0, the SOC decay emissions model
Comments	

Data / Parameter	$E_{BBGB}^{[m]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions from below-ground biomass at the end of the current monitoring period
Equations	[F.31]
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques to quantify above ground biomass, application of a root-shoot ratio and a decay emissions model.
Frequency of monitoring/recording	Every monitoring period
Value applied	1,501,822
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root:shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	

Data / Parameter	$E_{BBGB}^{[m-1]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions from below-ground biomass at the beginning of the current monitoring period
Equations	[F.31]
Source of data	Measurement in the proxy area

Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques to quantify above ground biomass, application of a root-shoot ratio and a decay emissions model.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Calculation of above ground biomass using standard vegetation sampling techniques and application of an acceptable root:shoot ratio and application of equation F.10 of VM0009 V3.0, the decay emissions model
Comments	0 for the first monitoring period

Data / Parameter	$E_{BBM}^{[m]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions from biomass at the end of the current monitoring period
Equations	[F.16], [F.30], [F.52]
Source of data	Measurement in the proxy area
Description of measurement methods and procedures to be applied	See Appendix E
Frequency of monitoring/recording	Every monitoring period
Value applied	4,438,058
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	Annual monitoring of proxy area. See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Standard vegetation sampling techniques, quantification of deforestation in the reference area and application of equation F.4 and F.24, VM0009 V3.0
Comments	

Data / Parameter	$E_{BSOC}^{[m]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions from soil carbon at the end of the current monitoring period

Equations	[F.16], [F.26]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Standard soil sampling techniques in the proxy area, laboratory analysis and application of the soil decay model.
Frequency of monitoring/recording	At first verification
Value applied	0
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of soil carbon and application of equation F.9 and F.29, VM0009, V3.0
Comments	SOC not included in this current monitoring period. See section 2.2.

Data / Parameter	$E_{BSOC}^{[m-1]}$
Data unit	tCO_2e
Description	Cumulative baseline emissions from soil carbon at the beginning of the current monitoring period
Equations	[F.26]
Source of data	Measurement in the project area
Description of measurement methods and procedures to be applied	Standard soil sampling techniques in the proxy area, laboratory analysis and application of the soil decay model.
Frequency of monitoring/recording	At first verification
Value applied	0
Monitoring equipment	See Appendix P, standard plot sampling equipment
QA/QC procedures to be applied	See Appendix P for error correction procedures.
Purpose of data	Calculation of baseline emissions
Calculation method	Quantification of soil carbon and application of equation F.9 and F.29, VM0009, V3.0
Comments	First monitoring period

Data / Parameter	$E_{BA}^{[m]}$
Data unit	tCO_2e
Description	Cumulative emissions allocated to the buffer account at

	the end of the current monitoring period
Equations	[F.55]
Source of data	VCS Non-Permanence Risk Report, See Appendix H
Description of measurement methods and procedures to be applied	See Appendix H – standard VCS procedure used.
Frequency of monitoring/recording	Non-permanence risk revised at every baseline re-assessment.
Value applied	-437,125
Monitoring equipment	None
QA/QC procedures to be applied	Annual review of risk criteria and disturbance monitoring
Purpose of data	Calculation of emissions submitted to VCS buffer account
Calculation method	Procedure followed in VCS non-permanence risk tool and emissions allocated to buffer account subtracted from GERs.
Comments	

Data / Parameter	$E_L^{[m]}$
Data unit	tCO_2e
Description	Cumulative emissions from leakage at the end of the current monitoring period
Equations	[F.44]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	Leakage per VM0009 v3.0 for first monitoring period is 0

Data / Parameter	$E_L^{[m-1]}$

Data unit	tCO_2e
Description	Cumulative emissions from leakage at the beginning of the current monitoring period
Equations	[F.44]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	This is first monitoring period

Data / Parameter	$E_{L\Delta}^{[m]}$
Data unit	tCO_2e
Description	Change in emissions due to leakage
Equations	[F.53]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	This is first monitoring period

Data / Parameter	$E_{L\ ASF}^{[m]}$
Data unit	tCO_2e
Description	Cumulative emissions from activity-shifting leakage in forested strata at the end of the current monitoring period
Equations	[F.45]
Source of data	Measurements in the leakage area(s)
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	None
QA/QC procedures to be applied	See Appendix R, standard vegetation sampling equipment
Purpose of data	See Appendix R, all evidence of tree-felling by humans are geo- referenced and plots photographed
Calculation method	Application of equation F.44 and F.45 to field data collected.
Comments	This is first monitoring period

Data / Parameter	$E_{LME}^{[m]}$
Data unit	tCO_2e
Description	Cumulative emissions from market leakage at the end of the current monitoring period
Equations	[F.45]
Source of data	N/A
Description of measurement methods and procedures to be applied	Land-use by deforestation agents is subsistence and therefore all market leakage is captured by monitoring in activity shifting leakage areas.
Frequency of monitoring/recording	N/A
Value applied	0
Monitoring equipment	None
QA/QC procedures to be applied	N/A
Purpose of data	N/A
Calculation method	N/A
Comments	

Data / Parameter	$E_{P\Delta}^{[m]}$

Data unit	tCO_2e
Description	Change in project emissions
Equations	[F.53]
Source of data	Monitoring records for Forest Fire, Burning, logging, wood products, and natural disturbance events
Description of measurement methods and procedures to be applied	Standard vegetation sampling techniques and remote sensing to determine the extent of fires and other disturbances.
Frequency of monitoring/recording	Every monitoring period
Value applied	464,197
Monitoring equipment	Standard vegetation sampling equipment and satellite imagery.
QA/QC procedures to be applied	Remote sensing, patrols and field measurements
Purpose of data	Calculation of project emissions
Calculation method	Monitoring data from areas of Forest Fire, Burning, logging, wood products, and natural disturbance events and equation F.41 VM0009 V3.0.
Comments	

Data / Parameter	$E_{P\Delta BRN}^{[m]}$
Data unit	tCO_2e
Description	Cumulative project emissions due to burning at the end of the current monitoring period
Equations	[F.41]
Source of data	Monitoring plots in the project accounting area
Description of measurement methods and procedures to be applied	Application of equation F.42 using burning monitoring data as input.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	Standard vegetation sampling equipment
QA/QC procedures to be applied	Review of monitoring records
Purpose of data	Calculation of project emissions
Calculation method	Monitoring data from areas of Burning and equation F.42 VM0009 V3.0.
Comments	Biomass burning in the project accounting area is not a project activity

Data / Parameter	$E_{U}^{[m]}$
Data unit	tCO_2e
Description	Cumulative confidence deduction at the end of the current monitoring period
Equations	[F.55]
Source of data	N/A
Description of measurement methods and procedures to be applied	Application of equation F.57 which is the a linear combination of weighted standard errors of estimates from baseline emission models and carbon stock measurements.
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	N/A
QA/QC procedures to be applied	Review of calculations
Purpose of data	Calculation of project emissions
Calculation method	Equation F.57 from VM0009 V3.0
Comments	

Data / Parameter	$P_{L\ DEG}^{[m]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation at the end of the current monitoring period
Equations	[F.46], [F.47], [F.48], [F.49]
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	Every monitoring period
Value applied	0.0815
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	$P_{L\ DEG}^{[m=0]}$
Data unit	Proportion (unitless)
Description	Portion of leakage due to degradation prior to first verification event
Equations	[F.48]
Source of data	Monitoring in the leakage area
Description of measurement methods and procedures to be applied	Plot based sampling of leakage area and estimation of degradation.
Frequency of monitoring/recording	First monitoring period
Value applied	0.0815
Monitoring equipment	See Appendix R, standard vegetation sampling equipment
QA/QC procedures to be applied	See Appendix R, all evidence of tree-felling by humans are geo-referenced and plots photographed
Purpose of data	Calculation of leakage
Calculation method	Calculated using equation B.9, VM0009 V3.0. Minimum sample size determined using equation B.31, VM0009 V3.0
Comments	

Data / Parameter	$t^{[i-1]}$
Data unit	Days
Description	Time from the project start date to beginning of monitoring period i
Equations	[F.32], [F.33]
Source of data	Monitoring records
Description of measurement methods and procedures to be applied	NA
Frequency of monitoring/recording	Prior monitoring period
Value applied	0
Monitoring equipment	NA
QA/QC procedures to be applied	NA

Purpose of data	Calculation of baseline emissions
Calculation method	Count of days in project crediting period
Comments	

Data / Parameter	$t^{[m]}$
Data unit	Days
Description	Time from project start date to end of current monitoring period
Equations	[F.19], [F.20], [F.24], [F.21], [F.25], [F.27], [F.28], [F.32], [F.33], [F.36], [F.37], [F.38], [F.39], [F.40]
Source of data	Monitoring records
Description of measurement methods and procedures to be applied	NA
Frequency of monitoring/recording	Every monitoring period
Value applied	1340
Monitoring equipment	NA
QA/QC procedures to be applied	NA
Purpose of data	Calculation of baseline emissions
Calculation method	Deduct project start date from end of date of current monitoring period.
Comments	

Data / Parameter	$t^{[m-1]}$
Data unit	Days
Description	Time from project start date to beginning of current monitoring period
Equations	[F.10], [F.36]
Source of data	Monitoring records

Description of measurement methods and procedures to be applied	NA
Frequency of monitoring/recording	Prior monitoring period
Value applied	0
Monitoring equipment	NA
QA/QC procedures to be applied	NA
Purpose of data	Calculation of baseline emissions
Calculation method	Deduct project start date from end of date of previous monitoring period.
Comments	

Data / Parameter	$U_B^{[m]}$
Data unit	tCO2e
Description	Total uncertainty in proxy area carbon stock estimate
Equations	[F.57]
Source of data	Biomass and soil carbon inventory data, Appendix G
Description of measurement methods and procedures to be applied	Calculation of the standard error of the total of the baseline emissions model, proxy area carbon stocks and accounting area carbon stocks
Frequency of monitoring/recording	Every monitoring period
Value applied	42,966
Monitoring equipment	NA
QA/QC procedures to be applied	Review of monitoring records
Purpose of data	Calculation of baseline emissions
Calculation method	Equation B.10 VM0009 v3.0

Comments	

Data / Parameter	$U_{EM}^{[M]}$
Data unit	tCO2e
Description	Total uncertainty in Baseline Emissions Models
Equations	[F.57]
Source of data	Forest State Interpretation, Appendix O
Description of measurement methods and procedures to be applied	Calculation of the standard error of the total of the baseline emissions model, proxy area carbon stocks and accounting area carbon stocks
Frequency of monitoring/recording	Every baseline revision
Value applied	37195
Monitoring equipment	NA
QA/QC procedures to be applied	Review of monitoring records
Purpose of data	Calculation of baseline emissions
Calculation method	Equation B.14 VM0009 v3.0
Comments	After removing SOC baseline from monitoring period 1

Data / Parameter	$U_{P}^{[m]}$
Data unit	tCO2e
Description	Total uncertainty in project accounting area carbon stock estimate
Equations	[F.57]
Source of data	Biomass and soil carbon inventory data, Appendix G
Description of measurement methods and procedures to be applied	Calculation of the standard error of the total of the baseline emissions model, proxy area carbon stocks and accounting area carbon stocks
Frequency of monitoring/recording	Every monitoring period
Value applied	9,863,177

Monitoring equipment	NA
QA/QC procedures to be applied	Review of monitoring records
Purpose of data	Calculation of baseline emissions
Calculation method	Equation B.10 VM0009 v3.0
Comments	

3.3 Monitoring Plan

Table 8. Summary of monitoring activities.

Monitoring Activity	Timing	Methods
Trespassing and boundary transgressions	Monthly patrols backed up with aerial surveys at least once per dry season.	Scouts patrol the project are on an ongoing basis, it is their core duty
Leakage monitoring	Annually	Forest Inventory team following VM0009 and remeasuring 100% of leakage plots each year as per protocol in Appendix R.
Plot measurements	Annually	Forest Inventory Team following VM0009, remeasuring 20% of plots in project accounting and project proxy areas as per protocol in Appendix P.
Disturbance monitoring	One a year using satellite imagery from April-May	Remote sensing using Sentinel-2 or Landsat imagery. Followed up with aerial surveys and, or scout patrols.

PDR.122

The principle means of monitoring project emissions is with inventory plots in project accounting areas where above ground biomass is measured by recording tree dbh. Allometric equations are used to convert the measures to above and below ground biomass. Soil samples are also collected during the first monitoring period to establish the soil carbon stocks for the project accounting area. VM0009 requires that all such plots must be measured during the first monitoring period and then 20% measured during each subsequent monitoring period. After mapping the project accounting area described in section 3.2.1 of the project document, the project used gis software to generate 122 plots randomly (Figure 10) within the project accounting area polygon. The sample size was estimated based on the variance in biomass inventory plot data from BCP's other REDD project, the Lower Zambezi REDD project.

MR.93

Plot monitoring is conducted according to the BCP field sampling manual (Appendix P). Optimum plot size and minimum tree diameter were determined under BCP's other project, the Lower Zambezi REDD Project. Circular plots (12m radius) are used due to their simplicity, ease of layout and reduced measurement error. A randomly generated plot location is navigated-to using a handheld GPS or Android Smart Phone. When the plot location is reached, the centre is permanently marked with a steel rebar driven into the ground.

Trees are identified to species level as far as possible. The use of a generic allometric equation to calculate AGOT mass fortunately negated mistakes or uncertainties with tree species identification.

As explained above, plots are re-measured on a rotating basis, with 20% re-measured per year, however stock estimates are only updated every 5 years after all plots have been re-measured.

In addition to the inventory plots, human and natural disturbances in the project accounting area will be mapped using visual interpretation of Sentinel-2 and if needed Landsat data. This will be done once a year using the most cloud free imagery from April-May. This represents the end of the raining season and beginning of the dry season. This is generally the best time of year for mapping woodlands in Zambia as there is less cloud cover, trees generally still have leaves, and the under story grass has started to dry out providing better contrast between grassy and wooded areas. The woodlands in the project area are fire adapted and fires generally don't cause massive disturbance. April-May imagery helps avoid mistakenly mapping fire scars as disturbance, when the fire likely only burned the under story grass. With April-May imagery, only permanently damaging fires will be mapped as disturbance.

When significant disturbances are detected in an area not covered by an inventory plot, a polygon will be drawn around the area and the per hectare emissions factors determined from the plot inventory work will be applied to the area to calculate project emissions.

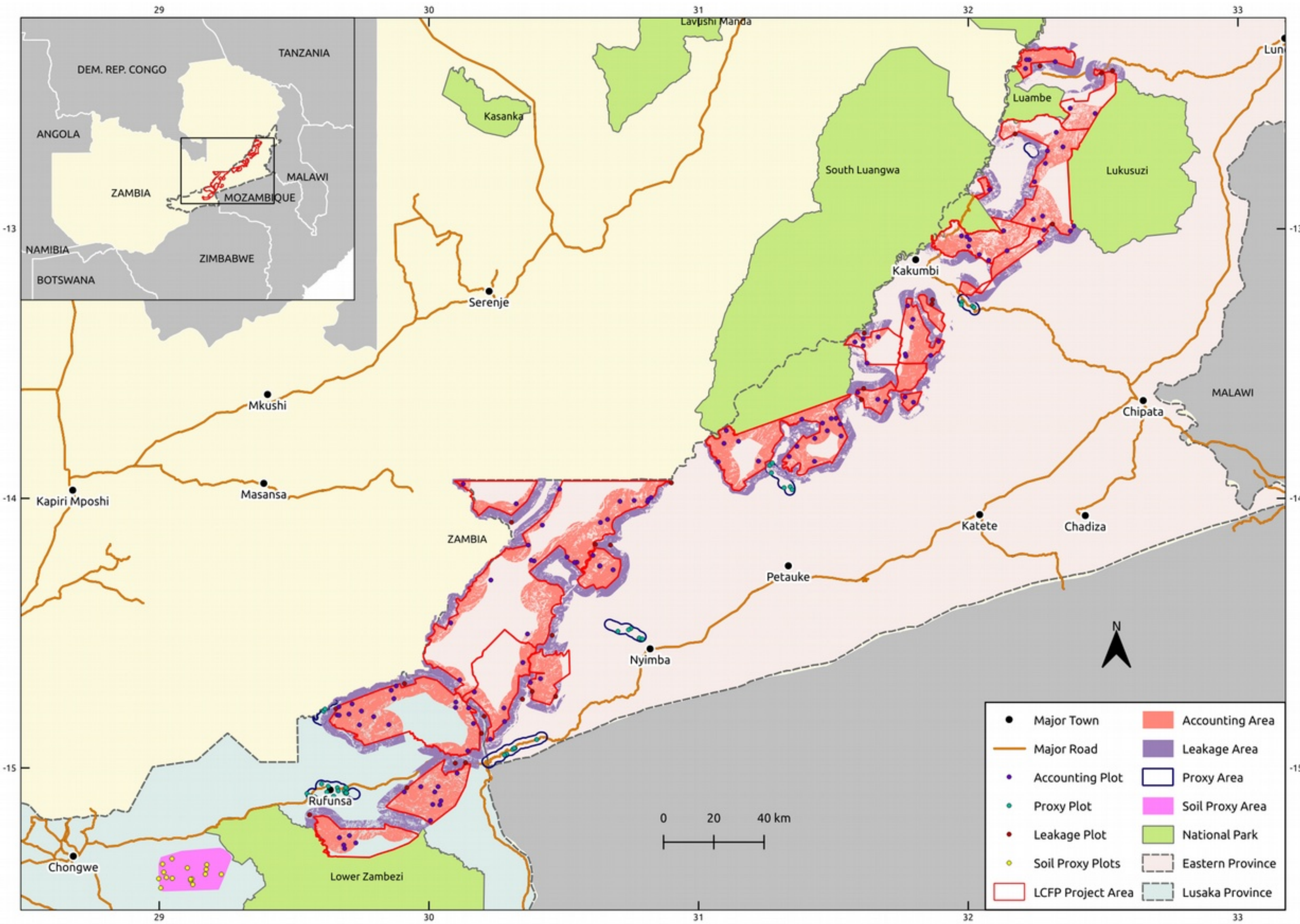

Figure 10: Accounting, Proxy, and Leakage Plots (**MR.91**)

PDR.123

Proxy plots are used to estimate residual biomass after conversion and the end point for soil carbon decay. For estimating residual biomass, after mapping the proxy areas as described in section 3.2.1 of the project document, the project generated a random sample of 45 points within the proxy areas. Variability in proxy plots is high since some farmers leave large isolated tress in their farms and invariability, some proxy plots will include there trees. Going based on experience from the previous LZRP REDD project, BCP settled on a sample size of just 45 points (Figure 10). This is because while the variability with proxy plots is high relative to the proxy plot average, that variability is low relative to the accounting area carbon stocks and thus represents a relatively small portion of the overall uncertainty in baseline emissions.

Since proxy soil carbon is only measured once, it is important to select samples from areas that represent the end point of conversion. This data was already available from the previous LZRP REDD project for 16 soil plots from older farms in that project's proxy area.

The same biomass and soil sampling protocols used for accounting areas (Appendix P) were used for proxy areas as well.

PDR.124

Leakage plots in the leakage area are used to estimate activity-shifting leakage from the project accounting areas. The project generated a random sample of 25 points within the leakage area where 1 ha leakage plots could be established (Figure 10). Equation B.31 as per VM0009, suggested the minimum number of plots required was 22.

The leakage plot protocol is described in Appendix R.

MR.92

GPS coordinates for all of the accounting area, proxy, and leakage plots can be found in Appendix I.

MR.98

All accounting area, proxy, and leakage plots were measured during monitoring period 1. The project will verify on an annual basis going forward and will therefore measure all leakage plots each year and 20% of accounting area and proxy plots each year.

Organizational Structure, responsibilities and competencies

BioCarbon Partners has a team of experts responsible for the carbon stock monitoring and standards compliance. As per VM0009, carbon stock monitoring is done using a combination of permanent plots in the project accounting area, proxy area, and leakage area, and remote sensing. Scout teams, are used to confirm encroachment and deforestation in the project area as part of the community engagement process to resolve encroachment (Figure 11). The carbon accounting department oversees all forest related data collection, processing, storage, GIS and remote sensing and electronic data capture systems with a central database (Figure 12).

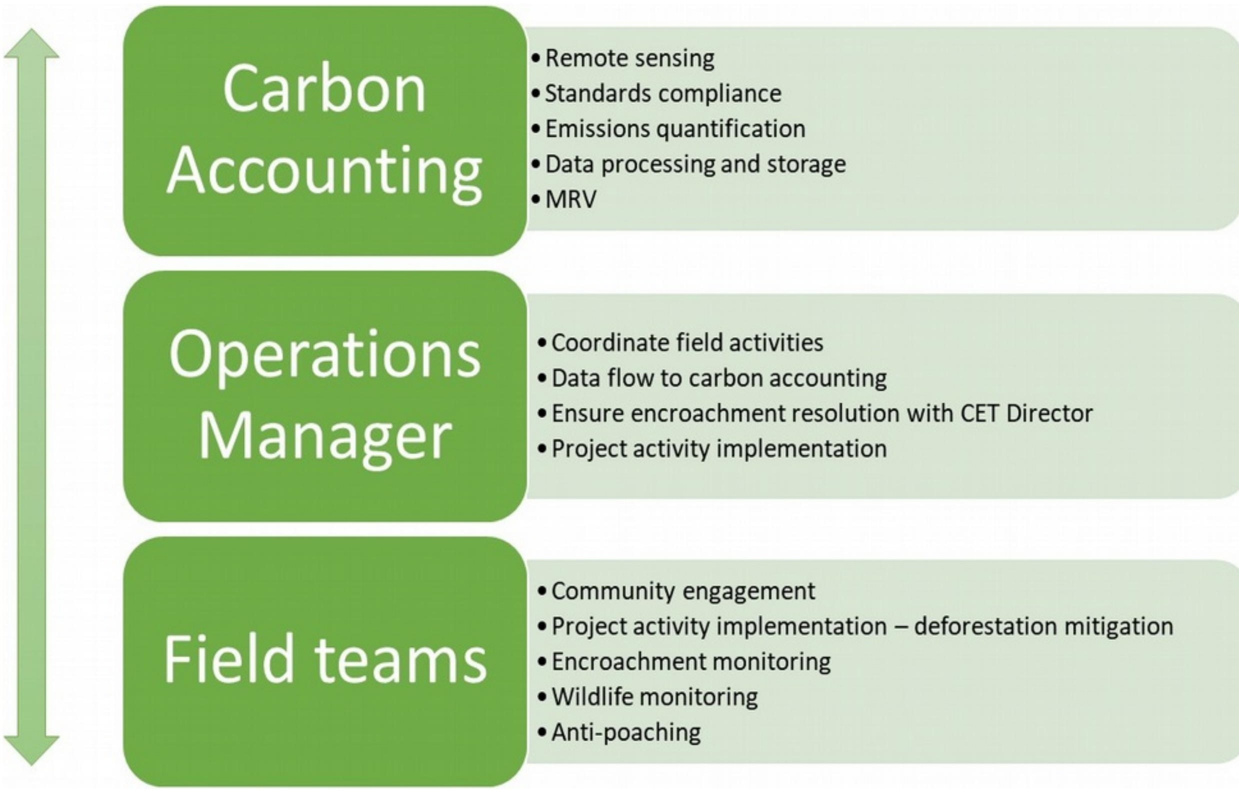

Figure 11. BCP Carbon Monitoring Process.

Both the carbon accounting manager and remote sensing and GIS specialist have post graduate degrees with a strong focus on African ecology and collectively have more than 20 years' experience in biological inventories, remote sensing and GIS.

Data Management

The process followed for field data collection is outlined in Figure 12. It is a feedback system that relies heavily on training, review and good communication. All training is followed by trial runs of the sampling procedure and then actual data collection under supervision. Data is continuously submitted from the field to the carbon accounting department, and feedback is provided if any corrective actions are required.

Remote Sensing data does not have a field component, but all data is also stored on Gdrive.

Once data has been error checked and cleaned it is safely stored online using Google Drive and then transferred to an online relational database.

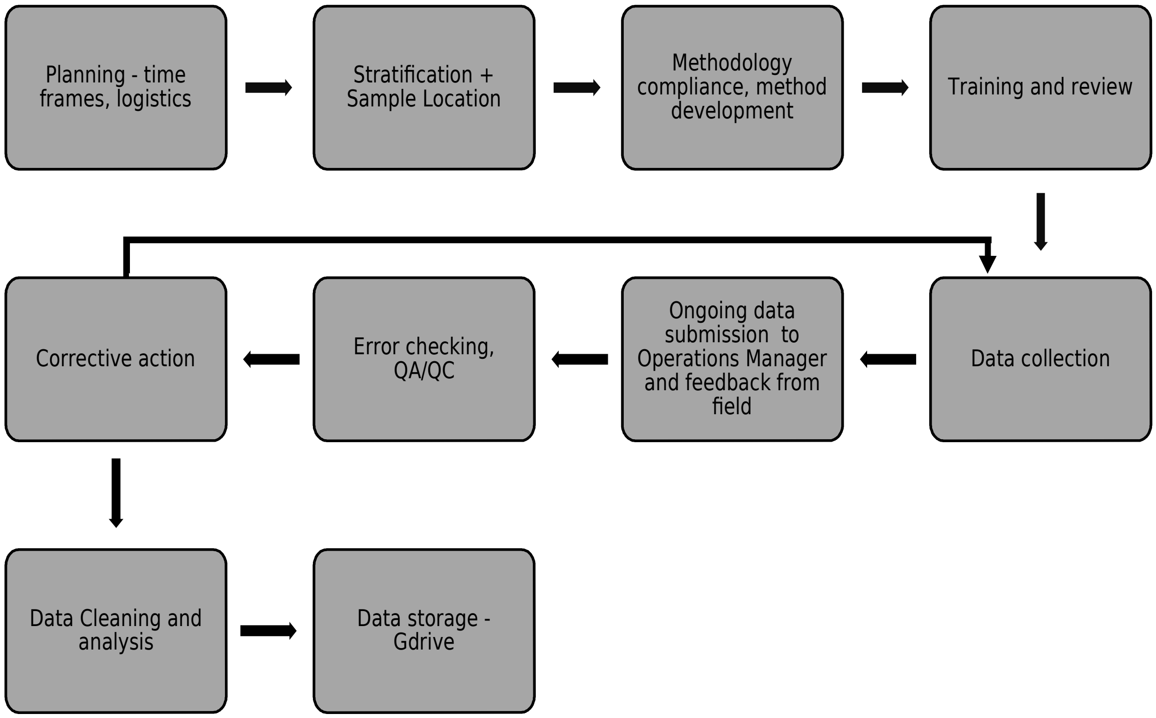

Figure 12. BioCarbon Partners data collection, QA/QC and storage process.

MR.90

Quality Assurance and Quality Control

All data submitted to the central database is subjected to quality control by appropriate staff. The majority of field data is collected using digital tools on smart phones and are either directly loaded to the database or saved on Gdrive where it is checked and then transferred to the database.

A set of Standard Operating Procedures and Field Manuals have been developed for all the various aspects of the carbon inventory. As indicated above, BCP has strong systems in place to ensure all carbon related data is collected according to the required quality standards. Quality Control is embedded in the organisational structure of the Carbon Accounting department (Figure 11).

Continuous feedback from the field, coupled with field visits from senior staff, ensure that errors are identified in a timely fashion and corrected (Figure 12). Spot checks and internal audits are conducted to make sure all data collected is of the highest standard. The new digital data capture system furthermore ensures that errors are identified and rectified immediately, unlike paper based systems where data often takes weeks to enter and error check. Spot errors are dealt with as per Appendix P.

With regards to GIS and RS, similar procedures are followed although this role resides mainly within the senior management's responsibility.

MR.88

Three forest monitoring teams were used to collect the biomass and SOC data. Each team was led by a experienced members of what was once a single team formed for the assessment of biomass and SOC data in the Lower Zambezi Redd+ Project. This initial team was trained over a period of 12 days in February and September or 2012. They received a refresher 2 day trainings in December of 2012, June of 2014, and May of 2016. This core team has been collecting data annually for the other REDD project since 2012 and are thus highly experienced.

In order to speed up the carbon inventory work, a second and then a third forest monitoring team was formed. Each of the new teams was lead by a highly experienced member of team 1 and all new team members received training before starting field work. Team 2 received a 2 day training including practical field experience in January 2019. Team 3 received the same two day training in May 2019.

MR.99

The project uses two allometric equations to estimate above ground biomass. For the majority of tree stems, a generic miombo equation developed by Chidymayo (2013) at a study site very close to the project area was used:

Wood Biomass (kg) = 0.0446 * $DBH^{2.765}$

It was based on destructive sampling of 113 trees of 19 different species (all of which are found in the project accounting area) with a DBH range of 2 to 39 cm.

The other equation used by the project is another general miombo equation developed by Mugasha et al, 2013 (see Appendix P for publication copy):

Wood Biomass (kg) = 0.1027 * $DBH^{2.4798}$

It was based on destructive sampling in 4 different sites in Tanzania of 167 sample trees of 60 tree species with a DBH range from 1.1 to 110 cm.

MR.100

Having been developed from samples trees in Zambia near the project area, the Chidumayo equation was the more local of the two equations. It also gave higher biomass estimates for all DBHs than the Mugasha equation. Therefore, it was the preferred equation. Unfortunately, it was not validated for trees above 55 cm. Therefore, it was applied to all trees with a DBH of 55 cm or less. It was also applied to trees less than 63 cm DBH, but the DBH of these trees was conservatively changed to be 55 cm since the equation had not been tested on trees larger than 55 cm DBH. This deviation is explained in more detail in section 2.2. The equation was used to estimate 80% of the project area carbon stocks.

For trees between 63 and 110 cm DBH, the Mugasha equation, which was validated for trees up to 110 cm DBH was applied. It was also applied to 3 trees with diameters greater than 110 cm DBH, by changing the DBH of these trees to 110 cm. This deviation is explained in more detail in section 2.2. The equation was used to estimate 20% of the project area carbon stocks.

MR.101

This is the first monitoring period. Both equations are used for this period as described.

MR.102

As per VM0009, there was no need to validate the Chidumayo equation with destructive harvesting because the equation came from existing published literature and was developed for miombo forests very similar to the project area. However, BCP did conduct some relevant destructive harvesting as part of developing it's first REDD+ project in Lusaka province. Additionally, BCP has the destructive harvesting data from Chidumayo's study of miombo biomass allometry in Lusaka province. So, these two datasets were used to validate the Mugasha et al. equation, which comes from further away.

Also, the samples used to develop both equations did not include trees as large as the largest trees in the inventory data and the equations do not pass the first order derivative test required by VM0009. Therefore, as described in MR.100 when the equations were applied to a tree with a DBH greater than the range used to develop the equation, the DBH was changed to match the maximum DBH for which the equation was validated. This deviation is described in more detail in section 2.2.

MR.103

Chidumayo (2013) developed the equation through destructive harvesting of study plots located between latitudes 15.2167° and 15.2833°S and longitudes 29.1500° and 29.2333°E, which was miombo woodland in Lusaka province. The sample consisted of 113 trees spanning 19 species, all of which occur in the project and proxy areas.

The Mugasha equation was also developed for miombo from 4 different sites in Tanzania. Out of 156 trees identified at the species level in the Mugasha et al. Dataset, 87 (56%) were species found in the accounting area biomass plots for this project. Out of 165 sample trees identified at the genus level, 141 (85%) were from a genus found in the accounting area inventory plots. Thus, there was a large amount of species and genus overlap between the project accounting area and the Mugasha et al. 2013 sample trees, showing that miombo in Tanzania is similar. Additionally, the equation appears to be conservative relative to the Chidumayo equation and predicts less biomass for any DBH than the Chidumayo equation.

MR.104

The Chidumayo equation did not require validation with destructive harvesting as it was developed from miombo sites in the center of Lusaka province within 60 km of the project area. Thus, the destructive harvesting data from the Chidumayo study and 7 additional trees destructively harvested by BCP were used the validate the Mugasha equation.

MR.105

Chidumayo (2013) developed the equation through destructive harvesting of study plots located between latitudes 15.2167° and 15.2833°S and longitudes 29.1500° and 29.2333°E, which was miombo woodland in Lusaka province within 60 km of the project accounting area. The diameter at breast height (DBH) range for the Chidumayo model was 2-39 cm and it was developed from a sample size of 113 destructively sampled trees covering 19 species, including most of the common species in the project accounting area. BCP then destructively harvested 7 more trees to bring the size range up to 55 cm DBH.

BCP has the original data from the author and have made it available to the VVB (Appendix G). As part of the work to establish it's first REDD+ project, BCP destructively sampled another seven trees of various species to extend the diameter range of the Chidumayo equation to 55cm (Table 12). The procedures followed for destructive harvesting are described in Appendix P and was developed under the guidance of Professor Chidumayo.

Species	DBH (CM)	Total dry mass (kg)	Mass Predicted by equation (kg)	Latitude	Longitude
Brachystegia boehmii	49	2750	2143	-15.3672	29.08735
Brachystegia boehmii	48	2838	1957	-15.3654	29.08534
Brachystegia boehmii	42	1236	1346	-15.3627	29.08307
Brachystegia spiciformis	41	1141	1318	-15.3678	29.08696
Brachystegia utilis	43	2212	1462	-15.3627	29.08266
Brachystegia utilis	48	2006	2030	-15.3635	29.08613
Parinari curatellifolia	55	3823	2950	-15.3654	29.08537
Total Mass		16006	13206		

MR.106

The field protocol is available in Appendix P.

MR.107

The field protocol (Appendix P) was based on the methods used by Chidumayo (2013) to develop the original equation. It is a methodical and robust approach that is conservative. The same wet-dry mass ratio as used by Chidumayo (2013) was applied.

MR.108

As shown in Appendix G, the Chidumayo equation predicts 10% less biomass than the cumulative measured biomass of the destructively sampled trees. Thus, it is acceptable under VM0009. The Mugasha equation is also acceptable under this condition, predicting 25% less biomass than the cumulative measured biomass of the destructively sampled trees (Appendix G).

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

MR.95

Table 9: Estimated carbon stock, standard error of the total for each stock, and the sample size for each stratum in the area selected.

Parameter	Project Accounting Area		Proxy Area	
	Forest (AGOT + BGOT)	Soil*	Proxy Area (AGOT + BGOT)	Proxy Area Soil*
Plots (P)	121	NA	45	NA
Total Possible plots ($N_{p,k}$)	42,927,183	NA	339,531	NA
C	114,668,494	NA	59,760	NA
U (B.10) per stratum	9,863,177	NA	15,411	NA

* SOC is excluded for this monitoring period. See section 2.2.

MR.10

Baseline emissions ($E^{[m]}_{(B\Delta)}$) for the first monitoring period (May 1st, 2015 to December 31st, 2018) are summarized in Table 10. Equation F.15 is used to calculate the baseline emissions for the current monitoring period. Cumulative baseline emissions were calculated using a logistic deforestation model, based upon a point interpretation of forest state in the reference area applying equation F.16 from VM0009 V3.0.

Table 10. Baseline emissions reduction summary for the sixth monitoring period (see Appendix G for calculations).

Monitoring Period	Carbon Pool	Baseline Emissions Reductions (tCO2e)
M1	AGOT	2,936,168
M1	BGOT	1,500,532
M1	SOC	NA
Total		4,436,700

MR.11

This is the first monitoring period so cumulative baseline emissions ($E_{(B\ \Delta)}^{[m-1]}$) from prior monitoring periods is 0.

MR.12

Cumulative baseline emissions for each selected pool ($E_{B\ BM}^{[m]}$ and $E_{B\ SOC}^{[m]}$) and undecayed carbon ($C_{B\ BGB}^{[m]}$, $C_{B\ SOC}^{[m]}$) as of the current monitoring period area summarized in Table 10. They are calculated using Equation F.16 from VM0009 V3.0. See Appendix G for calculations.

MR.15-17

Baseline emissions from biomass for the current monitoring period are based on the average carbon stock in selected carbon pools from AGOT, BGOT given by equation F.17 for the accounting area and F.18 for the proxy area. There is only one stratum with a carbon stock of 236.2 tCO2e. The proxy area (baseline scenario) carbon stocks are 15.6 tCO2e. Cumulative baseline emissions from biomass $E_{B\ BM}^{[m]}$ for this first monitoring period were calculated by equation F.24 from VM0009 V3.0. See Appendix G for calculations.

4,436,700 tCO2e

Current baseline emissions from SOC ($E_{(B\ \Delta\ SOC)}^{[m]}$) as of the current monitoring period would normally be calculated with equation F.29. See Appendix G for calculations. However, due to delays in SOC data, SOC is conservatively excluded for this monitoring period. See section 2.2 for more details.

Since this is the first monitoring period, cumulative baseline emissions from SOC ($E_{(B\ SOC)}^{[m]}$) would have been the same as current baseline emissions, which are conservatively set to 0 tCO2e for this monitoring period due to delays in SOC data.

MR.26

Carbon stored in non-decayed BGB ($C_{B\ BGB}^{[m]}$) for the current monitoring period was calculated using equation F.32. See Appendix G.

83,745 tCO2e

MR.27

Cumulative baseline emissions from BGB for the current monitoring period ($E_{B\ BGB}^{[m]}$) were calculated using equation F.30. See Appendix G.

1,585,568 tCO2e

MR.28

Since this is the first monitoring period, cumulative baseline emissions from BGB for prior monitoring periods ($E_{B\,BGB}^{[m]}$) is 0.

MR.29

Carbon stored in non-decayed SOC for the current monitoring period ($C_{B\,SOC}^{[m]}$) would have been calculated using equation F.33. However, due to delay in SOC data, SOC is excluded for this monitoring report (see section 2.2).

4.2 Project Emissions

MR.32

Major disturbances (generally, clearing for cropland preparation) in the project area were hand digitized from mostly cloud free Landsat 8 imagery from April-May 2015 and Sentinel-2 imagery from April-May, 2016 to April-May, 2018. Maps of the boundaries of these areas are shown in Figure 13, Figure 14, Figure 15, and Figure 16. The imagery was obtained from Earth Engine, exported to Google Cloud as map tiles (script provided to VVB), and then opened in QGIS where imagery from different years was compared to identify changes and polygons were drawn around the changes. The analysis somewhat exaggerates the amount of disturbance during the first period because it picks up on clearing that happened before the project start date, but which was not mapped by the remote sensing analysis with PALSAR data as non-forest. To reduce some of this error, the obviously cleared areas in May 2015, Landsat data that did not appear as non-forest in the PALSAR map were digitized and removed from the areas of detected deforestation. These areas of corrected error added up to 201.17 ha. Table 11 shows the breakdown by chiefdom and conservancy.

Table 11: Results of disturbance mapping in project area

Area Name	Clearing May 2015 – May 2018 (ha)
Chitungulu	1.34
Jumbe	0.04
Luembe	23.17
Malama	0.00
Mnkhanya	910.82
Mpanshya	46.80
Msoro	1,008.40
Munyamadzi	0.07
Mwanya	0.52

Nsefu	0.00
Nyalugwe	2.25
Pia Manzi	0.00
Sandwe	103.26
Shikabeta	3.54
Total	2,100.21

MR.33

The disturbed stratum represented just 0.43% of the project accounting area. By random chance, one of the original permanent sample plots fell into the disturbed area and was measured after disturbance. Therefore, in order to have sufficient plots to calculate variance, one more random plot location was selected inside the disturbed area and measured as per Appendix P. The disturbance stratum plots locations are show in Figure 15.

Current emissions from changes in project carbon stocks are calculated with equation F.41 VM0009 V3.0. After restratification (Table 12), the project accounting area stocks at the end of this monitoring period were equal to 114,204,406 tCO2e (Appendix G) compared to 114,668,494 tCO2e at the project start date. Thus project emissions for this monitoring period were -464,088 tCO2e.

Table 12: Estimated carbon stock, standard error of the total for each stock, and the sample size for each stratum in the project area at end of monitoring period.

Parameter	Project Accounting Area	
	Forest Statum (AGOT + BGOT)	Disturbed Stratum (AGOT + BGOT
Plots (P)	121	2
Total Possible plots ($N_{p,k}$)	42,723,653	203,530
C	114,172,448	31,958
U (B.10) per stratum	9,820,510	31,958

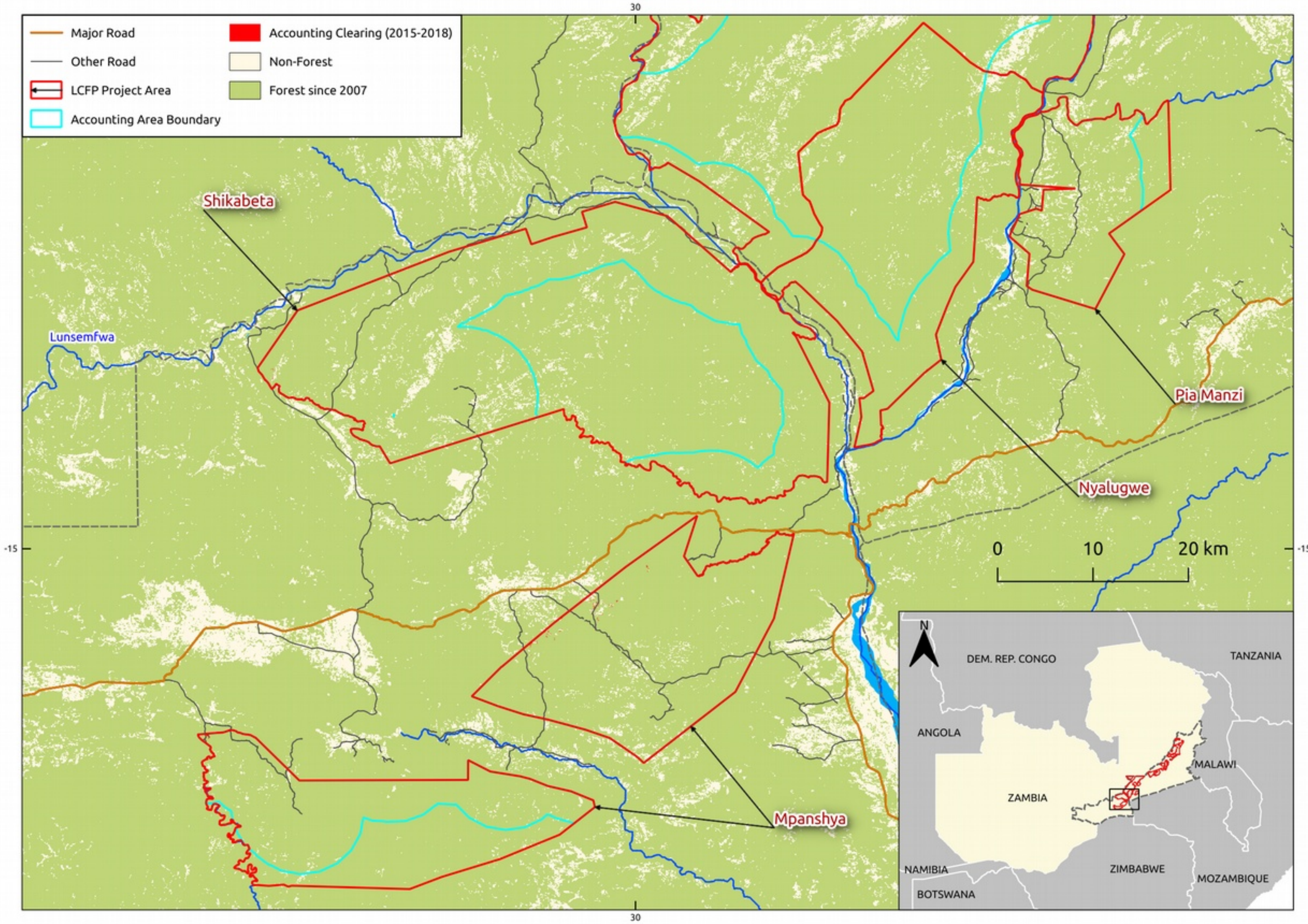

Figure 13: Significant disturbances in Mpanshya, Shikabeta, Nyalugwe, and Pia Manzi Accounting Areas

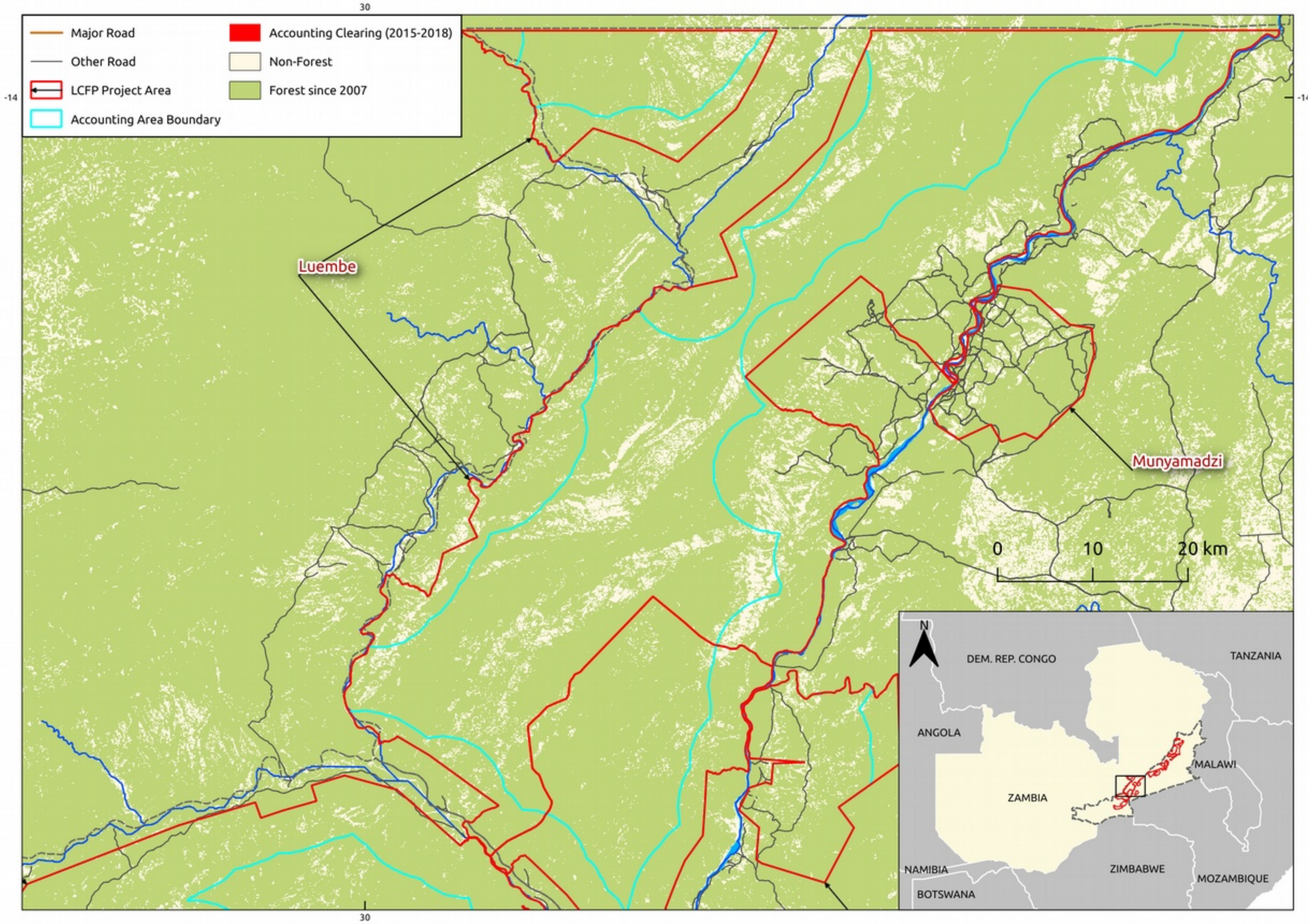

Figure 14: Significant disturbance in Luembe and Munyamadzi Accounting Areas

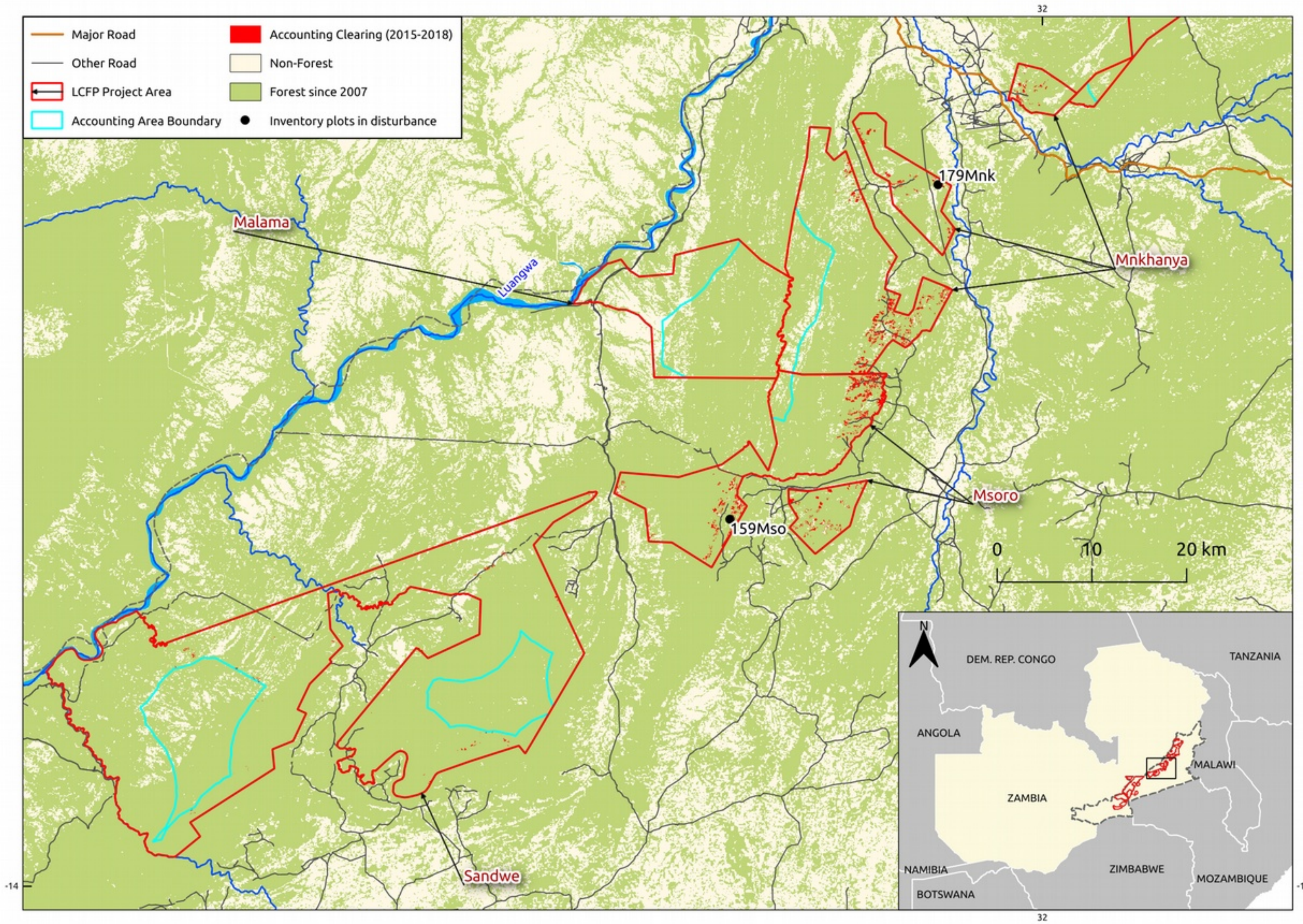

Figure 15: Significant disturbance in Sandwe, Malama, Msoro, and Mnkhanya Accounting Areas and disturbance inventory plots

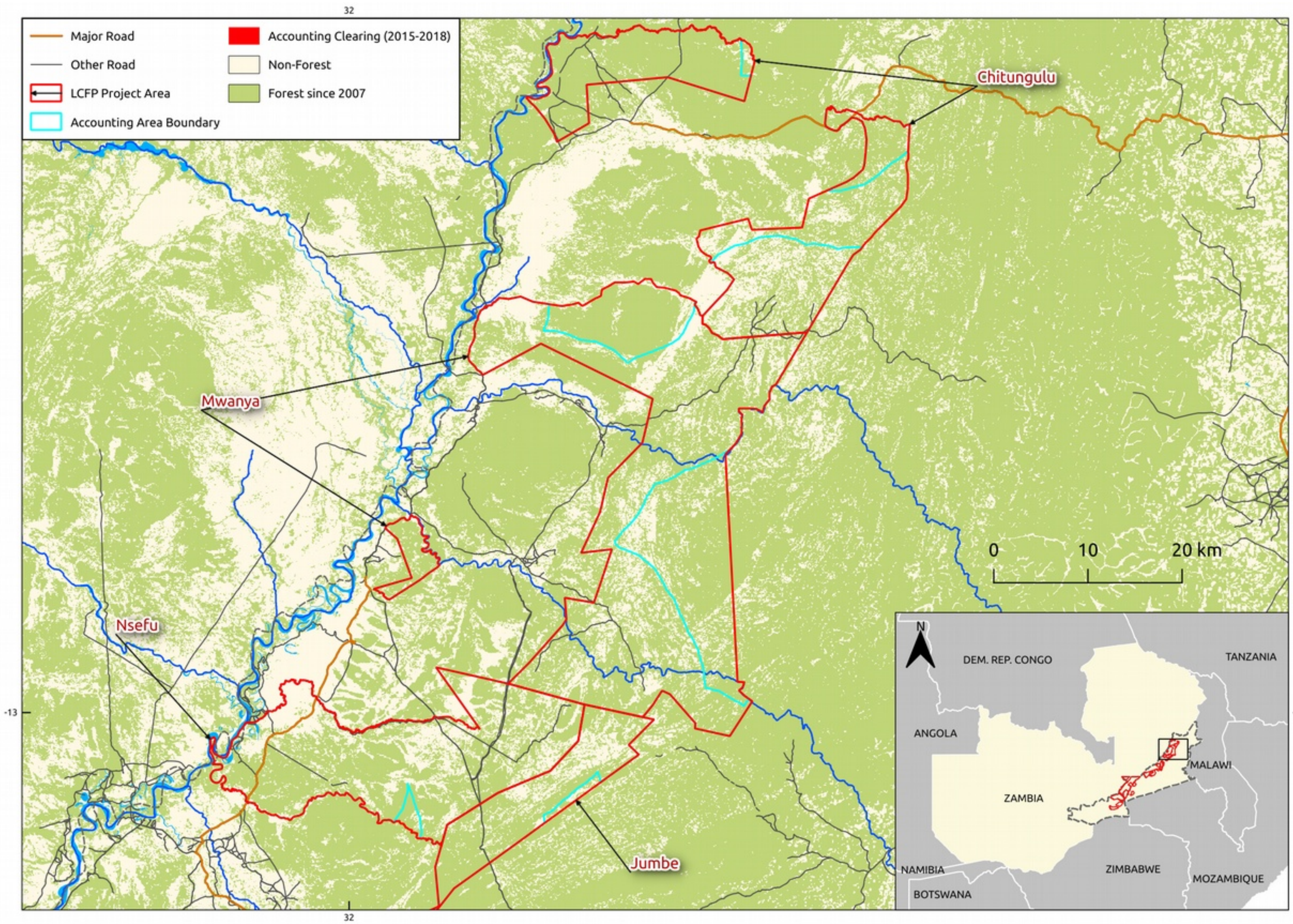

Figure 16: Significant disturbances in Jumbe, Nsefu, Myanya, and Chitulungu Accounting Areas

4.3 Leakage

MR.44

The primary leakage mitigation strategies of the project are improving agricultural techniques with conservation farming and agroforestry, and promoting alternative livelihoods that don't depend on forest clearing such as beekeeping. See section 2.1 activity area 2 of this report for more detailed descriptions of the leakage mitigation activities employed by this project. Based on experience of BCP's other REDD+ project in the Lusaka Province, which employed similar leakage mitigation activities and has not experienced any leakage in 10 years, it is assumed that these activities will also be effective at preventing leakage from the LCFP project area.

MR.48

Under VM0009 V3.0, for the first monitoring period, cumulative emissions from activity-shifting leakage ($E^{[m]}_{(L\ ASF)}$) are 0.

MR.49

This is the first monitoring period, so cumulative emissions from activity-shifting for the prior monitoring periods ($E^{[m]}_{(L\ AS)}$) are 0.

MR.56

The parameter ($p^{[m]}_{(L\ \ DEG)}$) for the current monitoring period (1) was estimated using equation B.9. The calculations can be found in Appendix G.

$p^{[1]}_{(L\ \ DEG)}$ = 0.0815

MR.57

This is the first monitoring period, so $p^{[1]}_{(L\ \ DEG)}$ = 0.0815

MR.63

BCP is a project proponent representing the private land-owners in this project. BCP doesn't own any land, but is the project proponent in one other REDD+ project in the country involving a private land-owner. The management plans and land-use of land in that REDD+ project have not changed as a result of this project.

The community resource boards (CRBs) are the project proponents for the project areas under community forest management agreements (CFMAs). The project areas are the only lands under their control.

4.4 Net GHG Emission Reductions and Removals

MR.65

GERs (Net GHG emissions or removals in Table 13) were calculated with equation F.53 as per section 8.4.1 of VM0009 V3.0 (see Appendix G).

MR.66

This is the first monitoring period so there is no prior monitoring period

MR.67

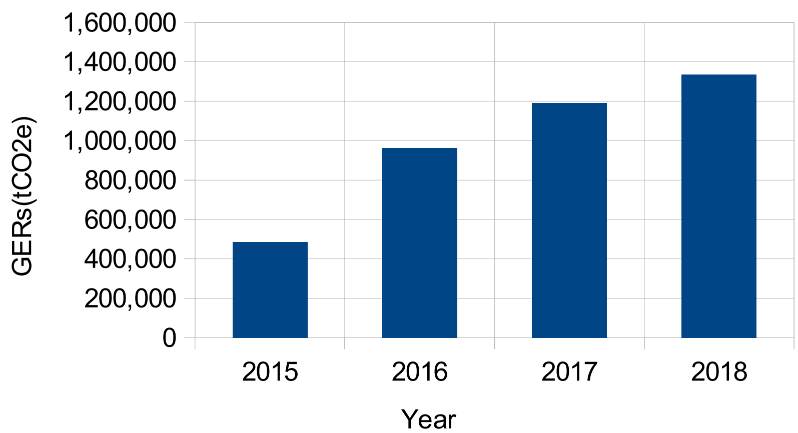

Figure 17: Graph of GERs by monitoring year.

Table 13: Net GHG emissions reductions or removals for current monitoring period

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)

2015	570,024	-84,506	0	485,518
2016	1,088,751	-126,758	0	961,993
2017	1,317,124	-126,412	0	1,190,712
2018	1,460,802	-126,412	0	1,334,390
Total	**4,436,701**	-464,088	0	3,972,613

MR.68

$U_B^{[m]}$ - Total uncertainty in proxy area carbon stock estimate = 42,966 tCO2e

$U_{EM}^{[m]}$ - Total uncertainty in Baseline Emissions Models = 37,184 tCO2e

$U_P^{[m]}$ - Total uncertainty in project accounting area carbon stock estimate = 9,863,177 tCO2e

$E_u^{[m]}$ - Total uncertainty = -118,622 tCO23 = 0

There was no confidence deduction.

MR.69

$U_B^{[m]}$ was calculated with equation B.10 VM0009 V3.0

$U_{EM}^{[m]}$ was calculated with equation F.14 VM0009 V3.0

$U_P^{[m]}$ was calculated with equation B.10 VM0009 V3.0

$E_u^{[m]}$ was calculated with equation F.57 VM0009 V3.0

For more details see Appendix G.

MR.74

NERs (VCUs as per VCS) generated for the current monitoring period were calculated with equation F.55 from VM0009 V3.0 and are reported in Table 14.

Table 14: NERs (VCUs) for the current monitoring period.

Year	Current Baseline Emissions (tCO2e)	Current Project Emissions (tCO2e)	Current Leakage (tCO2e)	NERs (tCO2e)	Buffer account (tCO2e)	VCUs (tCO2e)
2015	570,024	-84,506	0	485,518	-53,407	432,111
2016	1,088,751	-126,758	0	961,993	-105,819	856,173
2017	1,317,124	-126,412	0	1,190,712	-130,978	1,059,733
2018	1,460,802	-126,412	0	1,334,390	-146,783	1,187,607
Total	**4,436,701**	**-464,088**	**0**	**3,972,613**	**-436,987**	**3,535,624**

MR.75

This is the first monitoring period so there are no previous NERs

MR.76

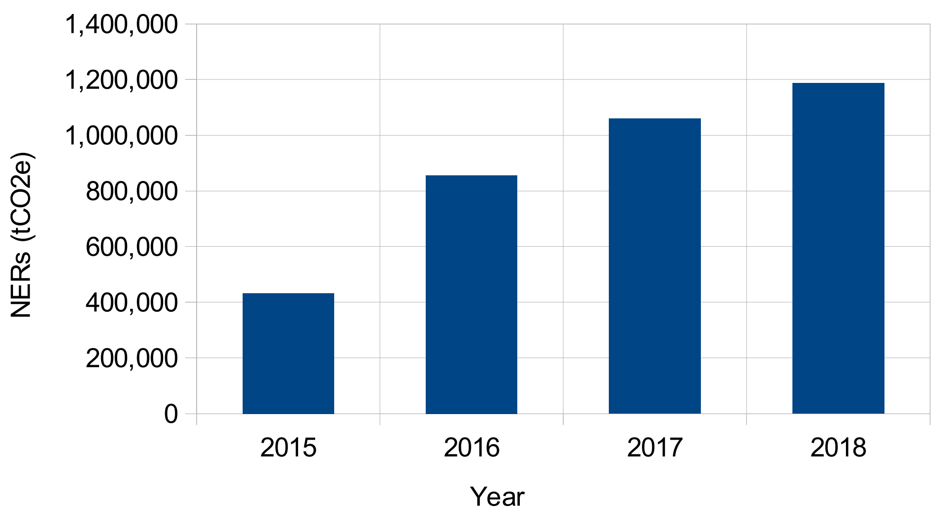

Figure 18: Graph of NERs by monitoring year.

MR.77

VCS Non-Permanence Risk Assessment Tool V3.2 (Appendix H) was used to calculate buffer account allocation and calculated by GERs x Non Permanence Risk score (11%) = -436,987 tCO2e.

MR.78

Please see Appendix H for non-permanence risk calculations using the VCS Non-Permanence Risk Assessment Tool V3.2.

MR.82

See Table 14 for quantified NERs by vintage year for the current monitoring period. See Appendix G for calculations.

MR.83

The ex-ante estimated NER for this monitoring period was 3,820,081 tCO2e compared to an ex-post estimate of 3,535,624.

MR.84

The difference between ex-ante and ex-post estimates is primarily due the exclusion of SOC emissions for this monitoring period since soil data was not yet complete at verification.

APPENDICES:

Appendix A Community Forest Management Agreements and Communication Agreement. Confidential

Appendix B Approval letters. Confidential

Appendix C Support letters. Confidential.

Appendix D Adaptive Management Plan. Confidential

Appendix E BCP Community Engagement SOPs and Monitoring Systems. Confidential

Appendix F Socio-economic baseline study. Confidential

Appendix G Emissions Reductions Calculations. Confidential

Appendix H VCS Non-permanence risk assessment. Public

Appendix I Spatial Data. Confidential

Appendix J Biodiversity Monitoring Results

Appendix K Analysis of agents, drivers, and underlying causes of deforestation. Internal Report. Confidential.

Appendix M Human Resource Manual and Grievance Procedure. Confidential

Appendix O Cumulative Deforestation Model. Confidential

Appendix P Biomass and Soil Sampling Protocol. Confidential

Appendix R Leakage Monitoring Protocol. Confidential